Programmer-Transparent Near-Data Processing Using Multiple Compute-Capable Resources in Solid-State Drives

Rakesh Nadig

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6th Workshop on Memory-Centric Computing Systems

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SAFARI

ETH zürich

Introduction

- **Rakesh Nadig**
  - PhD Student @ SAFARI research group since 2021
  - Previously worked at Samsung Semiconductor Research
  - MS in ECE from University of California Irvine

- **Research Interests**
  - Computer Architecture
  - Storage Systems
  - NAND Flash Memory
  - Near-Data Processing
  - Machine Learning
  - Heterogeneous Memory & Storage Systems

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Conduit

Programmer-Transparent Near-Data Processing Using Multiple Compute-Capable Resources in Solid State Drives

Rakesh Nadig, Vamanan Arulchelvan, Mayank Kabra, Harshita Gupta, Rahul Bera, Nika Mansouri Ghiasi, Nanditha Rao, Qingcai Jiang, Andreas Kosmas Kakolyris, Yu Liang, Mohammad Sadrosadati, and Onur Mutlu

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Inria

Talk Outline

Background and Motivation

Conduit

Evaluation

Summary

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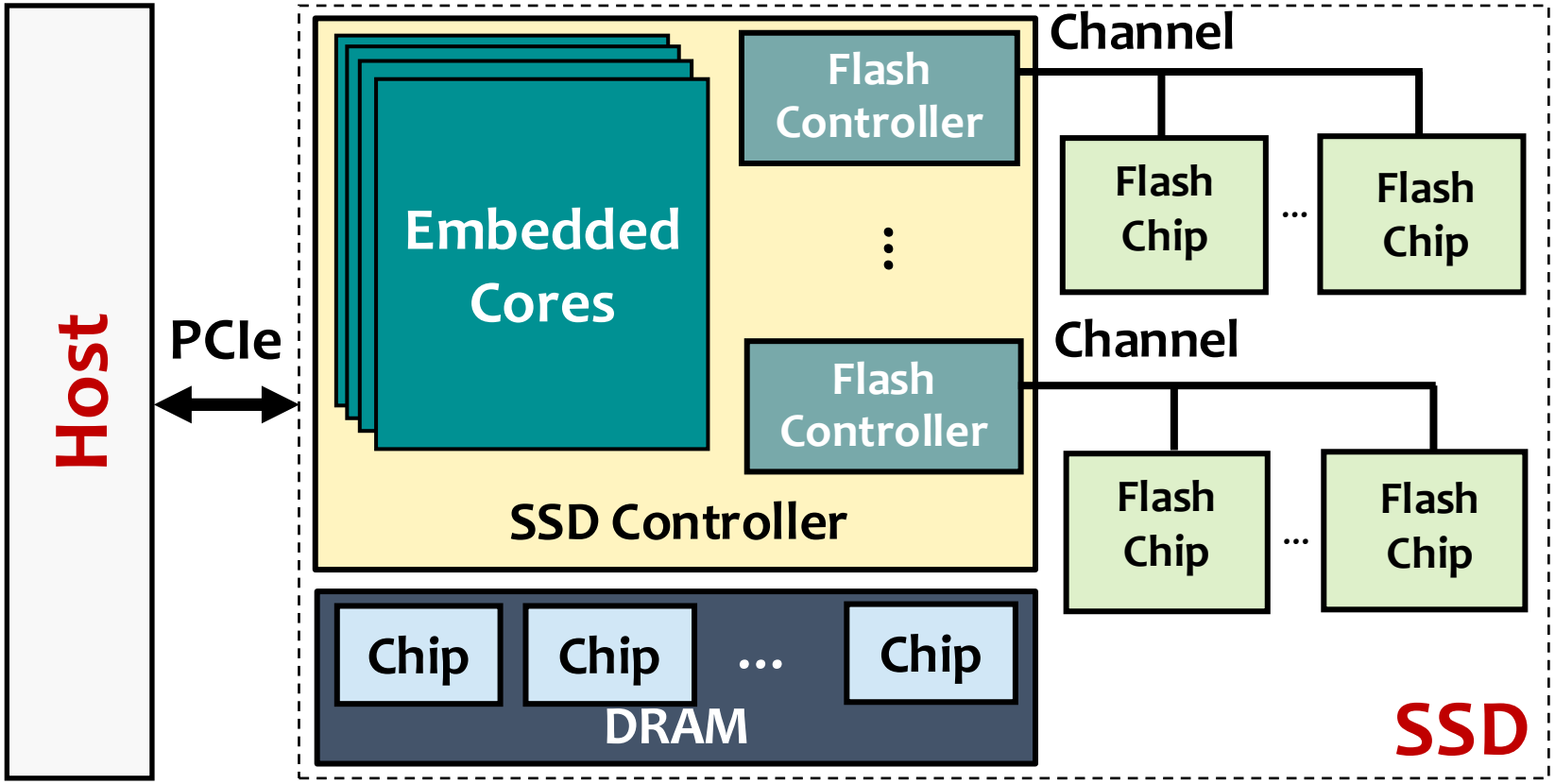
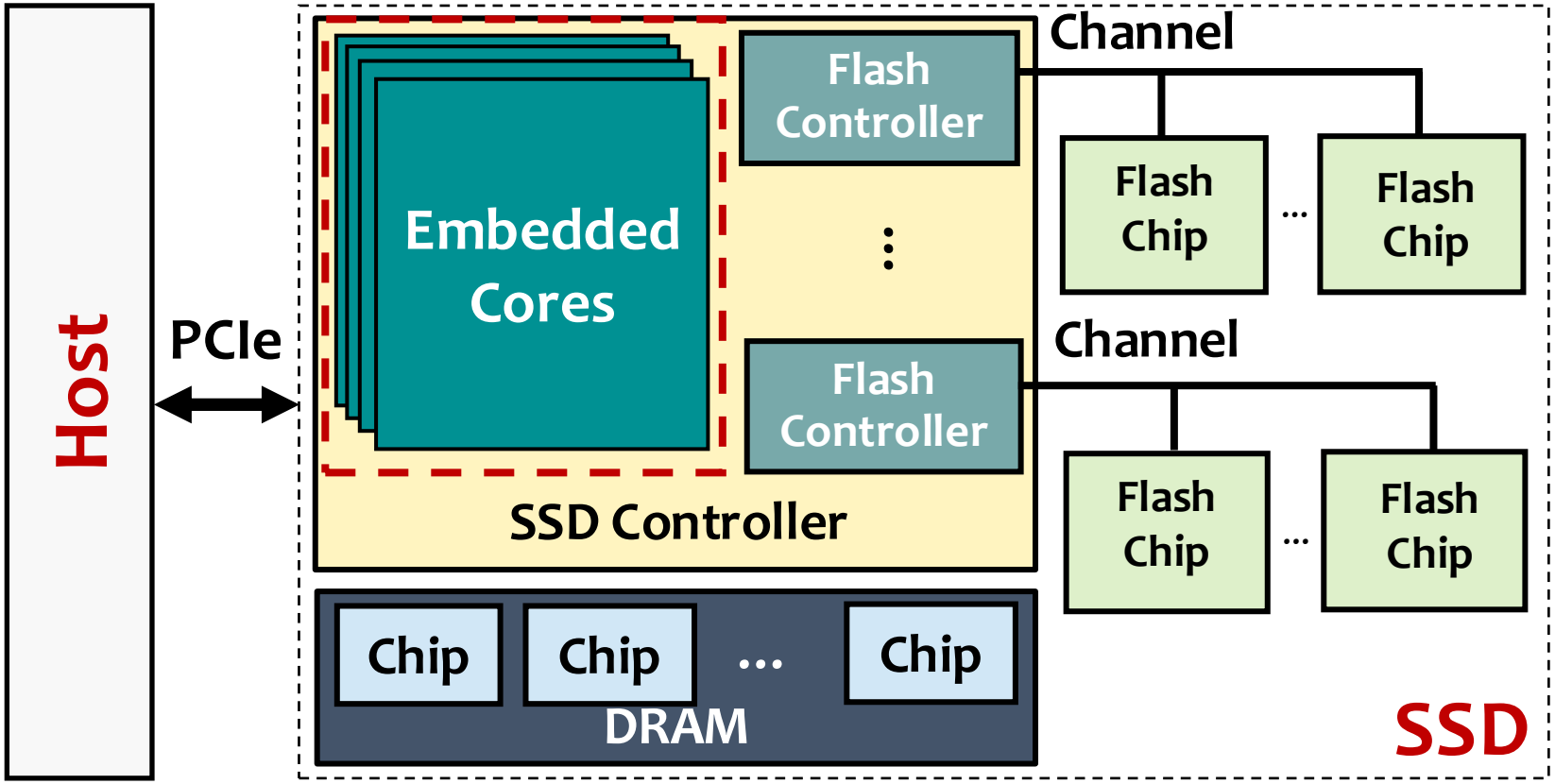
Overview of a Modern SSD



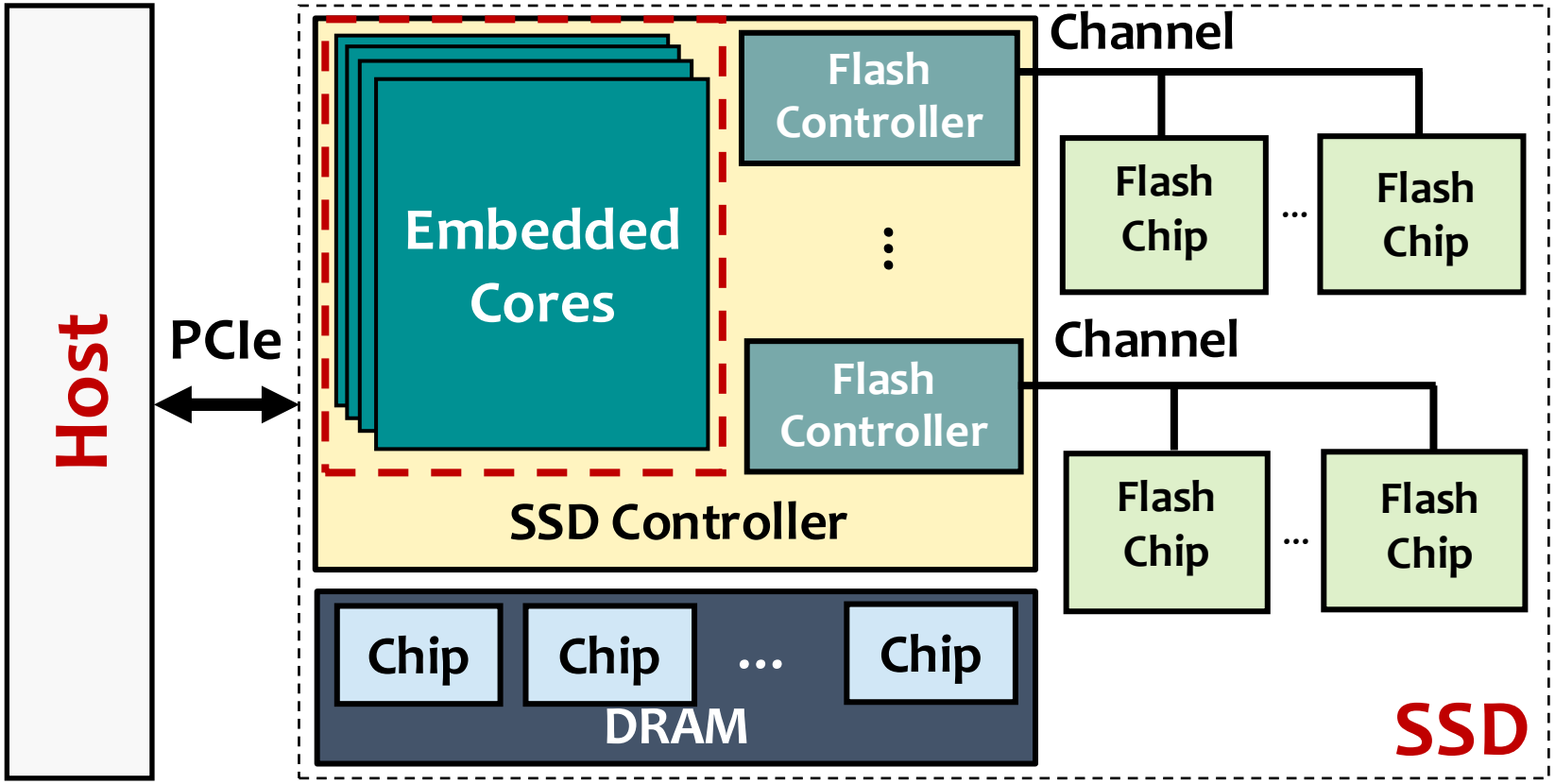
Near Data Processing in an SSD (ISP)



In-Storage Processing (ISP)

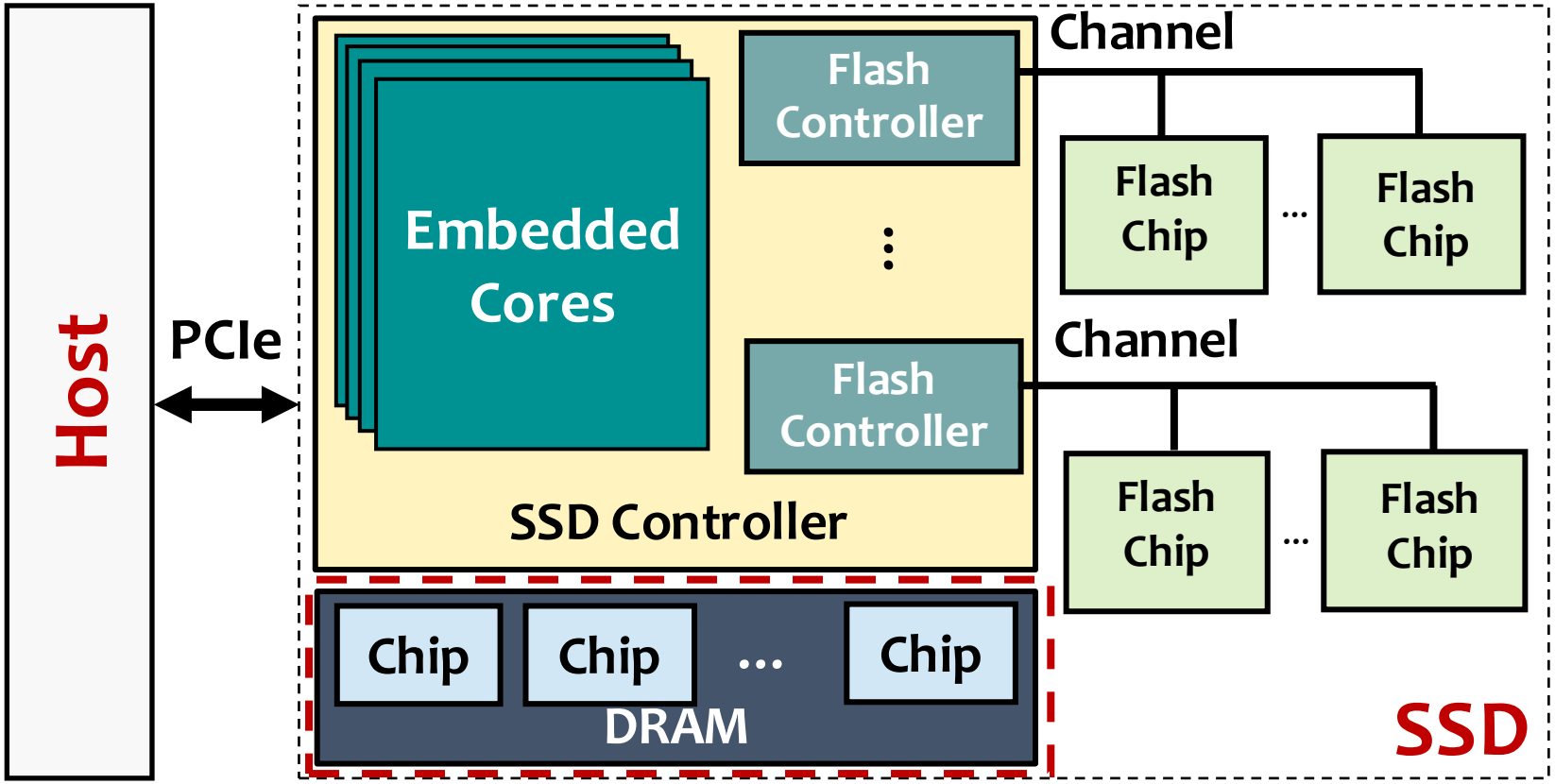
Processing using embedded cores in the SSD controller

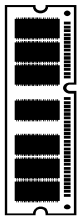
Near Data Processing in an SSD (ISP)



Many prior ISP works (e.g., Biscuit [Gu+, ISCA 2016], GenStore [Mansouri Ghiasi+, ASPLOS 2022], REIS [Chen+, ISCA 2025]) accelerate complex tasks including filtering, aggregation, encryption, and search

Near Data Processing in an SSD (PuD-SSD)

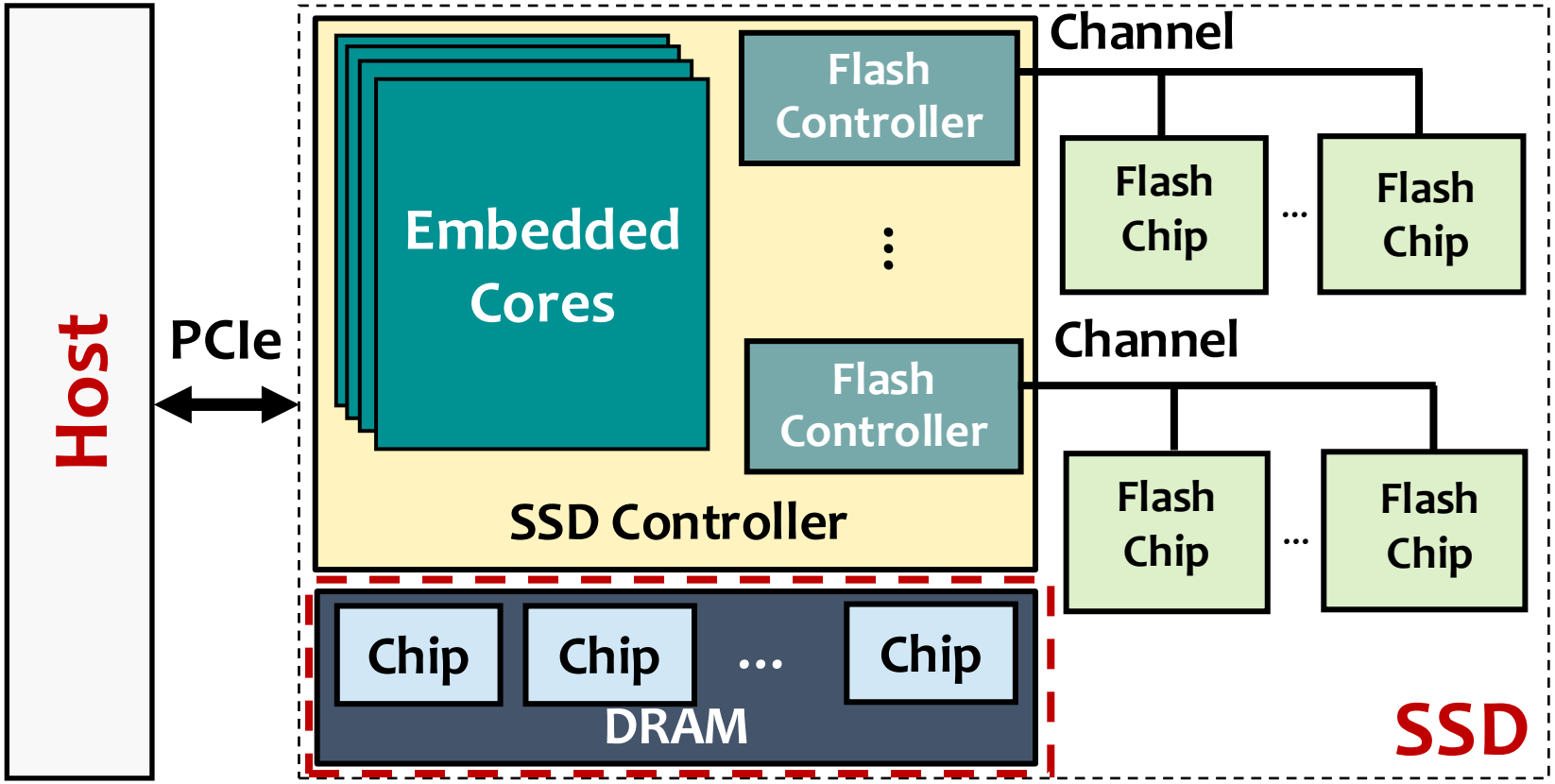




Processing-using-DRAM in SSD (PuD-SSD)

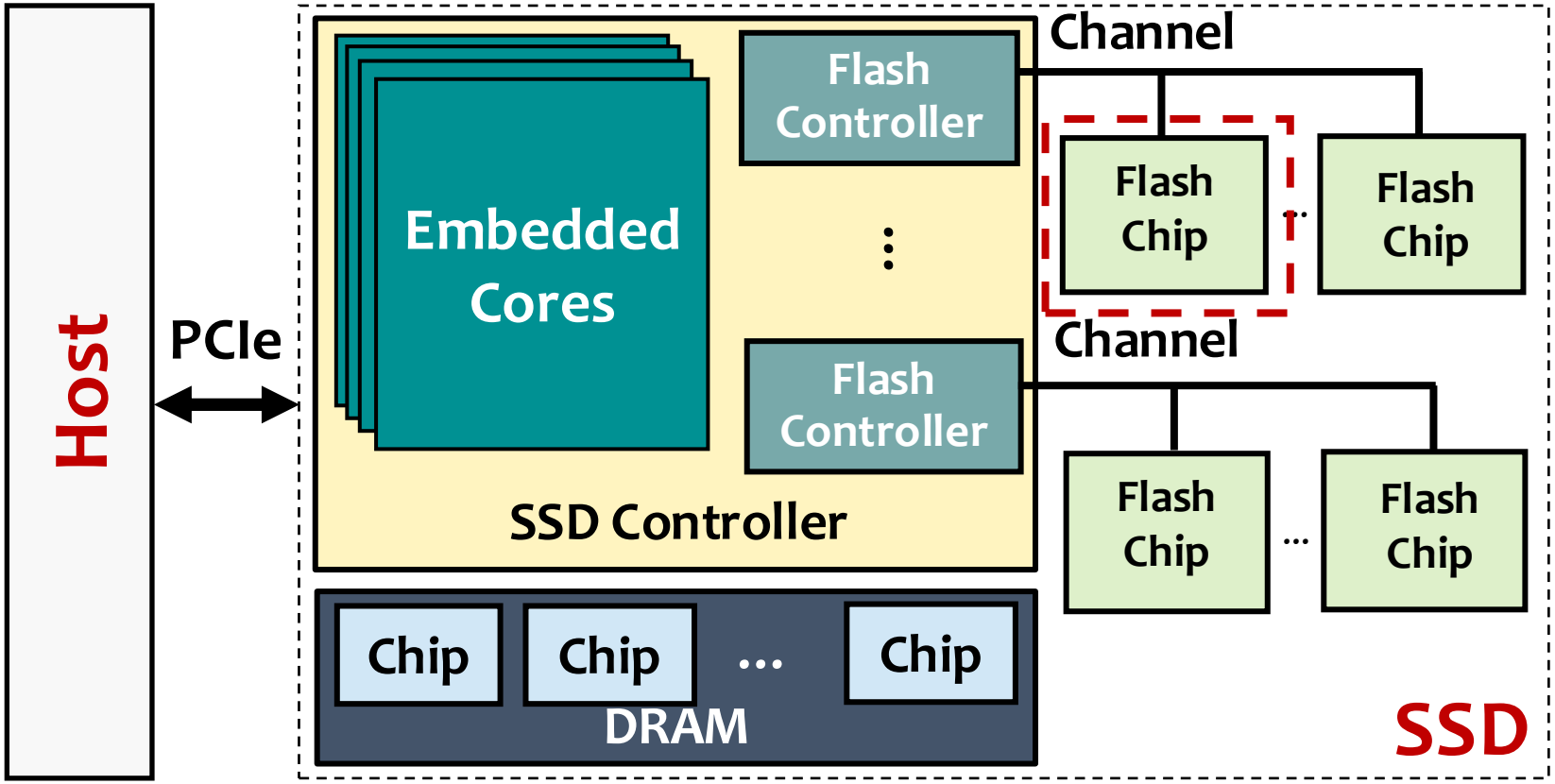
Computation using SSD DRAM's intrinsic operational principles

Near Data Processing in an SSD (PuD-SSD)



PuD works (e.g., Ambit [Seshadri+, MICRO 2017], SIMDRAM [Hajinazar+, ASPLOS 2021], MIMDRAM [Oliveira+, HPCA 2024]) enable bulk-bitwise and complex arithmetic operations entirely within the DRAM chips

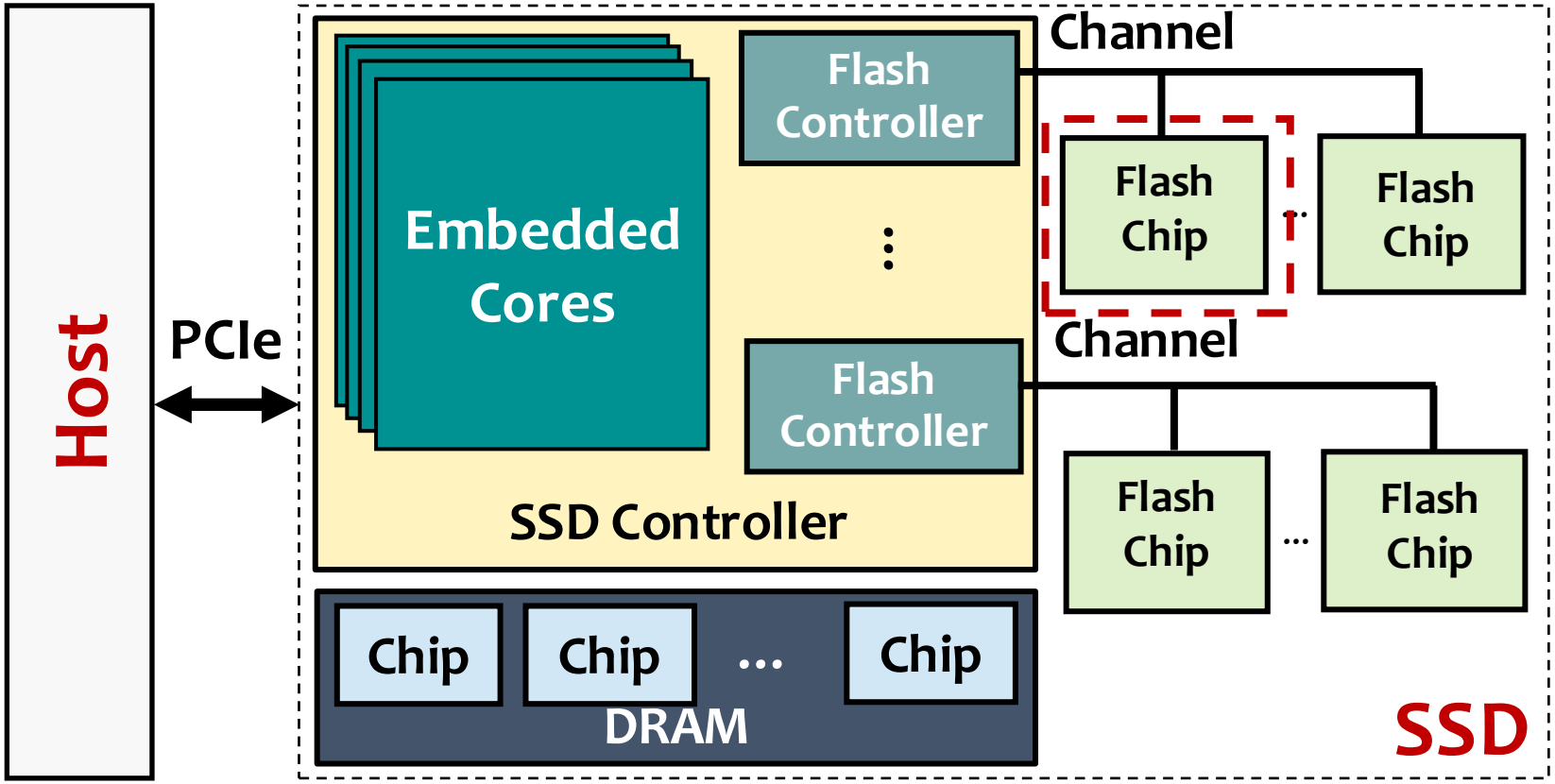
Near Data Processing in an SSD (IFP)



In-Flash Processing (IFP)

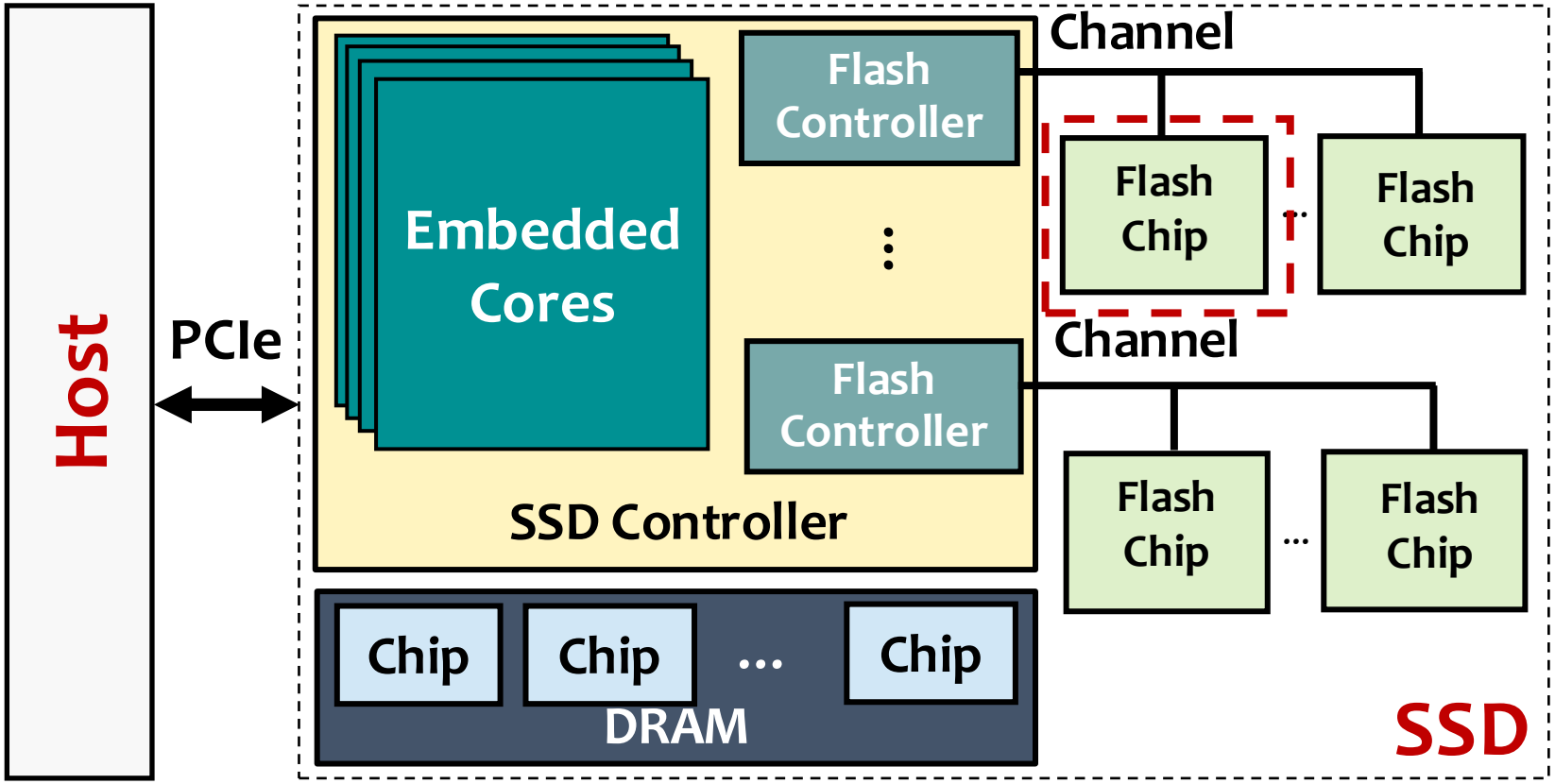
Computation by exploiting the flash chip's structure

Near Data Processing in an SSD (IFP)



Several IFP techniques (e.g., Ares-Flash [Chen+, MICRO 2024]) enable arithmetic operations inside NAND flash chips by controlling the latching circuit of the page buffer

Near Data Processing in an SSD (IFP)



Flash-Cosmos [Park+, MICRO 2022] enables bulk-bitwise operations within NAND flash chips by simultaneously reading multiple pages

Near Data Processing in an SSD: Tradeoffs

NDP Paradigm		
ISP	Supports general-purpose computations	Wimpy embedded cores provide limited performance
PuD-SSD	Computations in a highly parallel manner within the DRAM chips	Requires transferring operands from flash chips via bandwidth-limited flash channels
IFP	Eliminates data movement overhead and provides high computation parallelism	Limited operation set

NDP in an SSD: Research Questions

?

Is a single NDP paradigm good for all workloads?

?

Which is the most suitable NDP paradigm for each computation?

?

Do we need to combine these NDP paradigms?

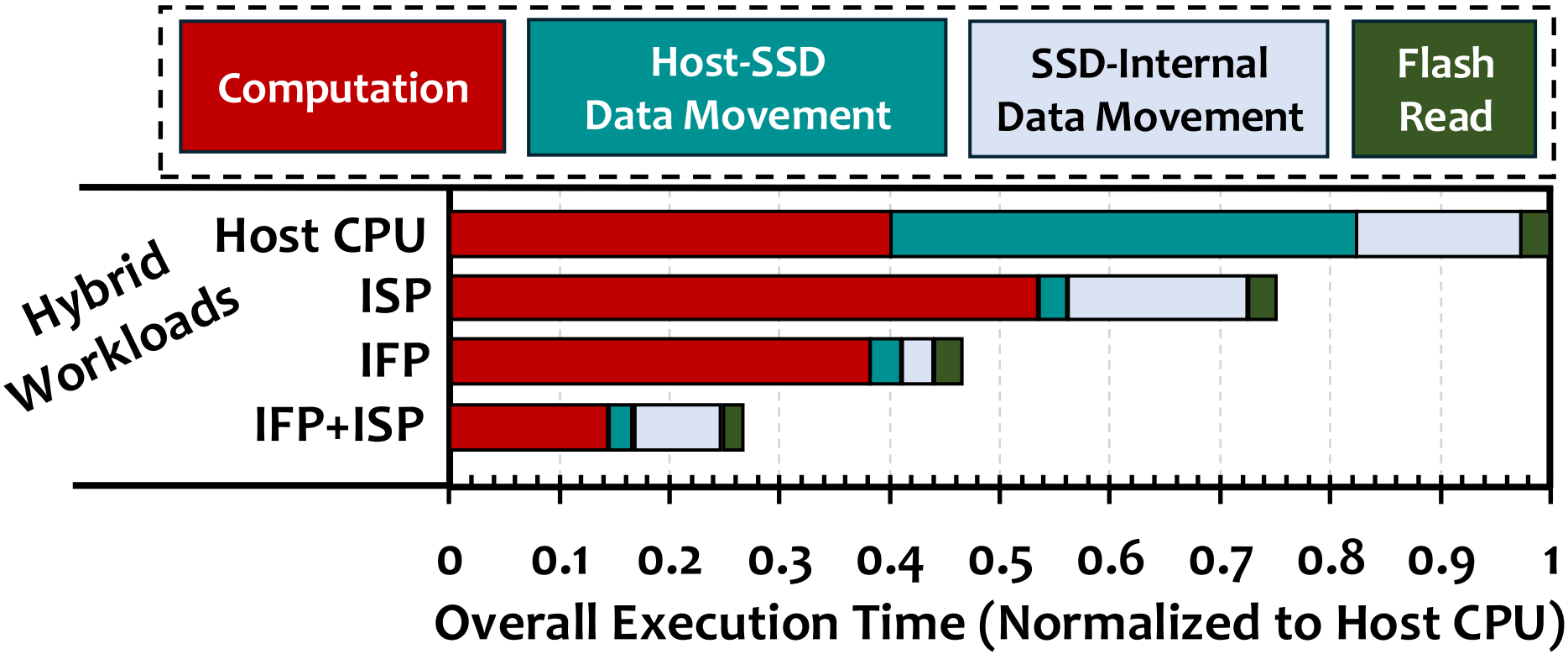
Case Study on Combining NDP Paradigms (1/5)

- We examine the performance bottlenecks by hand-mapping parts of the workload to multiple SSD computation resources

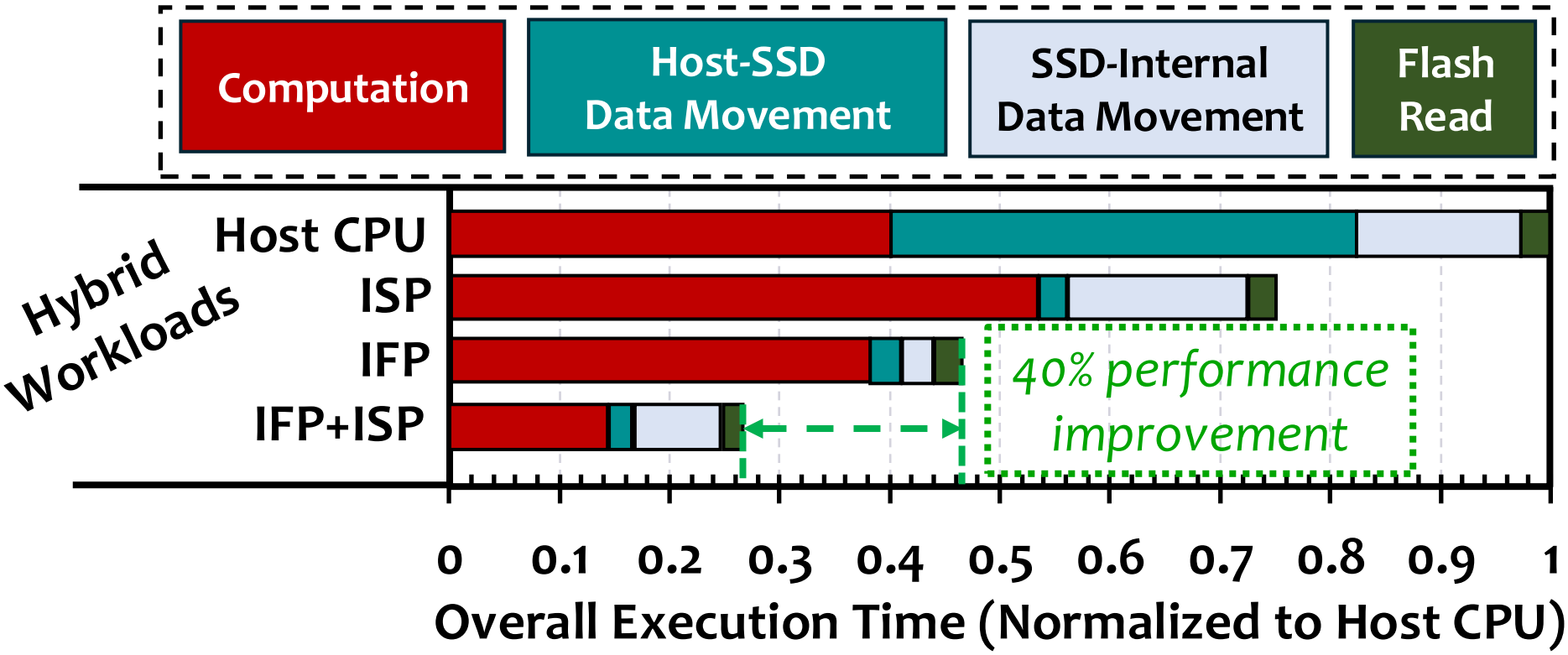
- Evaluated Policies
  - Host CPU
  - In-storage processing (ISP)
  - In-flash processing (IFP)
  - Offloading to both ISP and IFP (ISP + IFP)

- Evaluated Workloads
  - Hybrid workloads: Workloads that involve both computation and data movement (e.g., aggregation, sorting)
  - I/O-Intensive workloads: Workloads dominated by data movement (e.g., databases, bitmap indices)
  - More in the paper

Case Study on Combining NDP Paradigms (2/5)

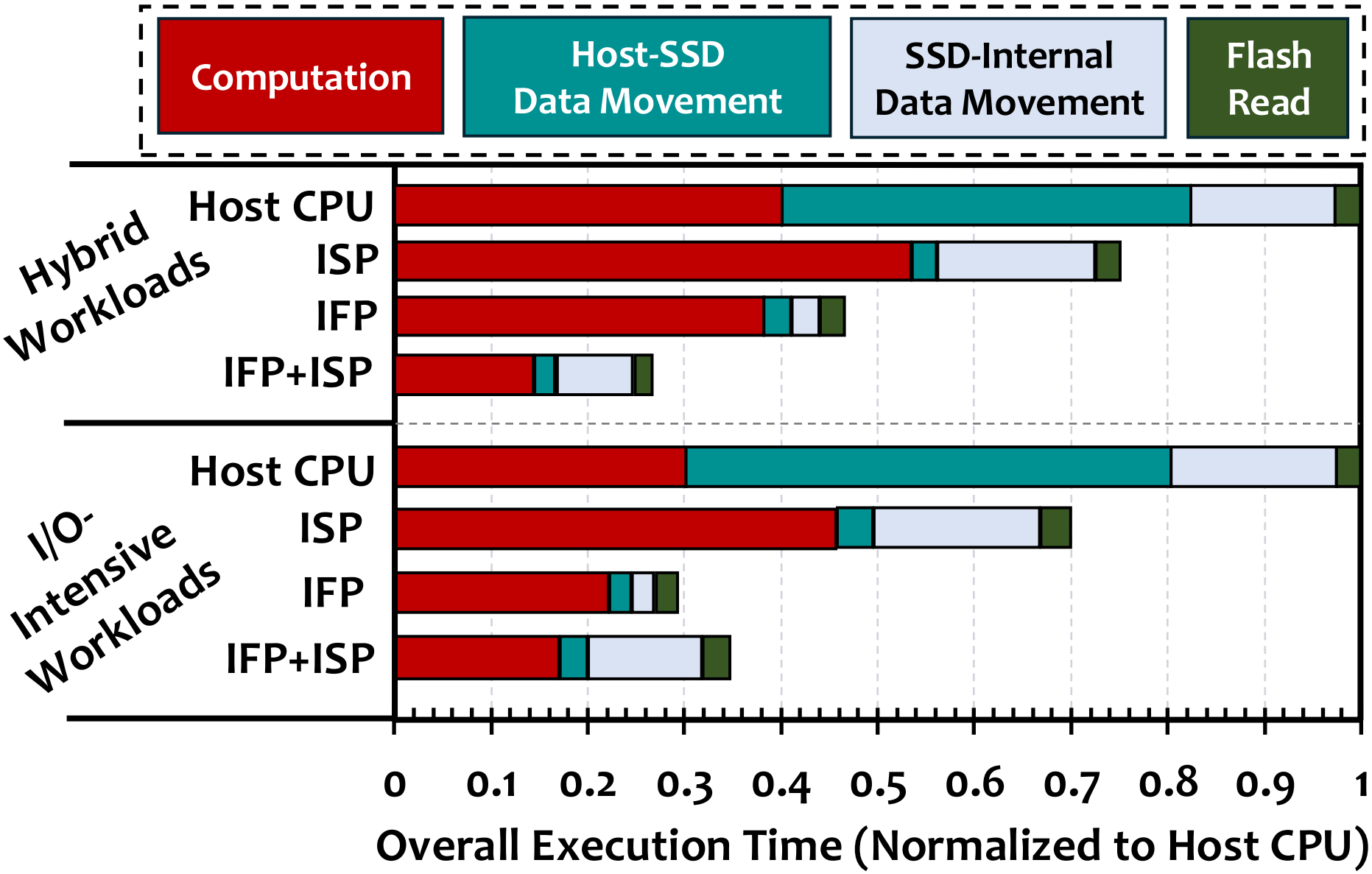


Case Study on Combining NDP Paradigms (3/5)

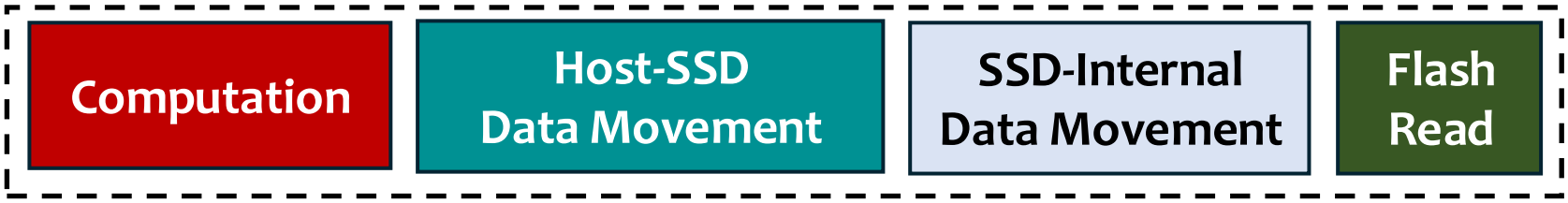


Offloading to ISP and IFP improves performance by 40% over IFP by reducing data movement and mapping computations to the most suitable computation resource

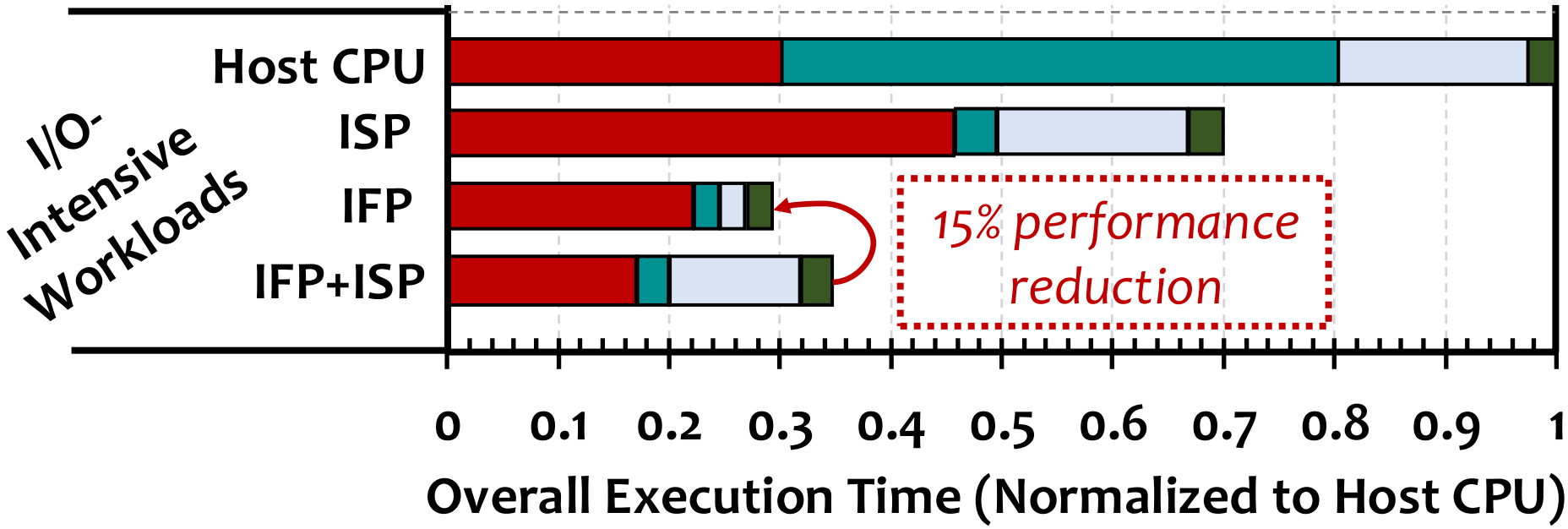
Case Study on Combining NDP Paradigms (4/5)



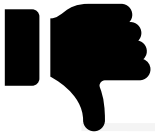
Case Study on Combining NDP Paradigms (5/5)



Naively combining multiple computation resources in I/O-intensive workloads negates offloading benefits due to data movement between computation resources



Case Study: Key Takeaways



No single computation resource consistently performs well across different workloads

Performance bottlenecks are workload-dependent and vary across different execution phases of the same workload

Offloading to multiple SSD computation resources provides benefits, but only when done judiciously

Limitations of Prior SSD-Based NDP Techniques

- Prior SSD-based NDP techniques

Map computations only to one or two NDP paradigms and fail to exploit the SSD's full computational potential

Lack generality across a wide range of applications

Are not programmer-transparent

NDP Offloading Techniques For Main Memory

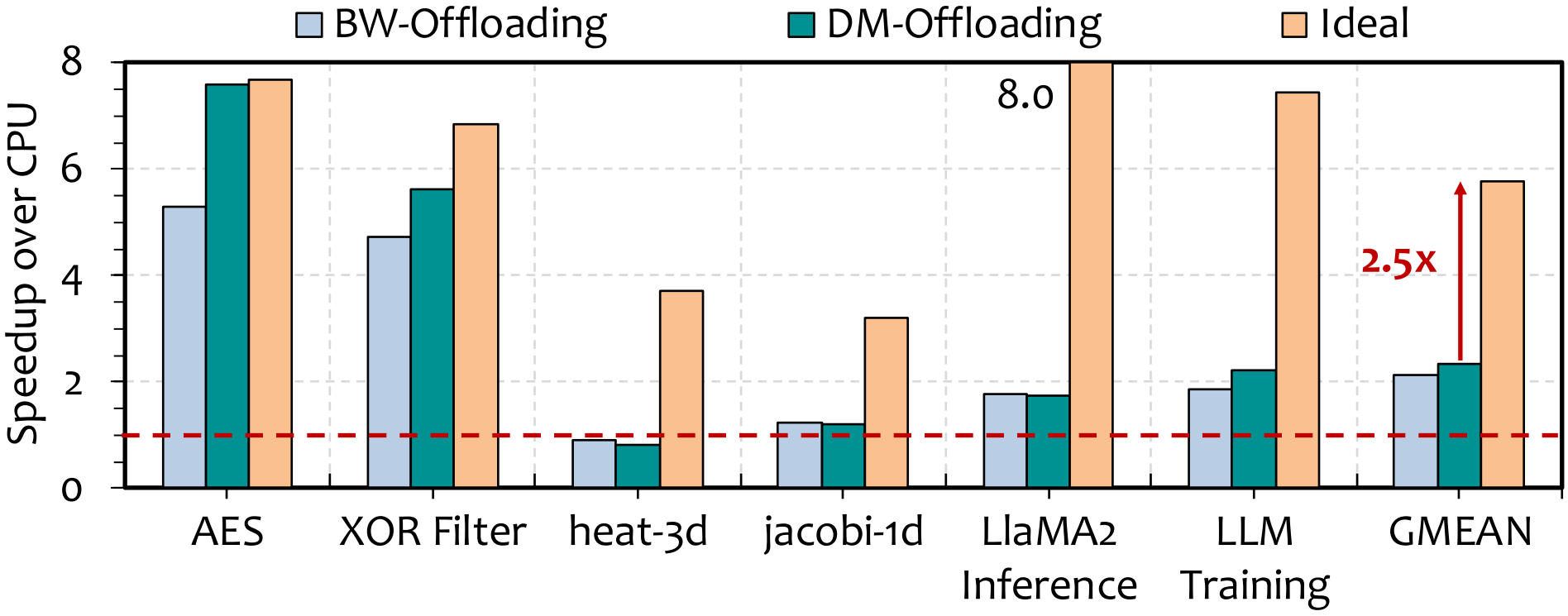
- No prior work proposes application-transparent offloading of computations across multiple SSD computation resources
- Several NDP offloading techniques explore partitioning and mapping application code segments between host and NDP units near main memory

BW-Offloading: Makes offloading decisions based on bandwidth utilization of the host and NDP units
(e.g., TOM [Hsieh+, ISCA 2016], ƛ-IO [Yang+, FAST 2023])

DM-Offloading: Makes offloading decisions to minimize data movement between the host and NDP units
(e.g., ALP [Mansouri Ghiasi+, TETC 2023], PIMProf [Wei+, DATE 2022])

- We adapt these offloading models for SSD-based NDP

Effectiveness of Prior Offloading Techniques



DM-Offloading falls short of the Ideal's performance by 2.5x due to its narrow offloading model

DM-Offloading prefers a single computation resource to reduce data movement, causing queueing delays and resource contention

Our Goal

To enable programmer-transparent NDP in an SSD that

- effectively utilizes multiple SSD computation resources to perform computations
- makes workload- and system-aware offloading decisions
- improves performance and energy efficiency of a wide range of applications

Talk Outline

Background and Motivation

Conduit

Evaluation

Summary

Our Proposal

Conduit

A general-purpose programmer-transparent NDP framework that dynamically offloads computations at instruction granularity to multiple heterogeneous SSD computation resources

Conduit: Key Steps

Compile-time pre-processing step to identify offloadable code regions in the application and transform them to wide vector operations

Runtime offloading of vectorized instructions to the most suitable SSD computation resource based on application and system characteristics

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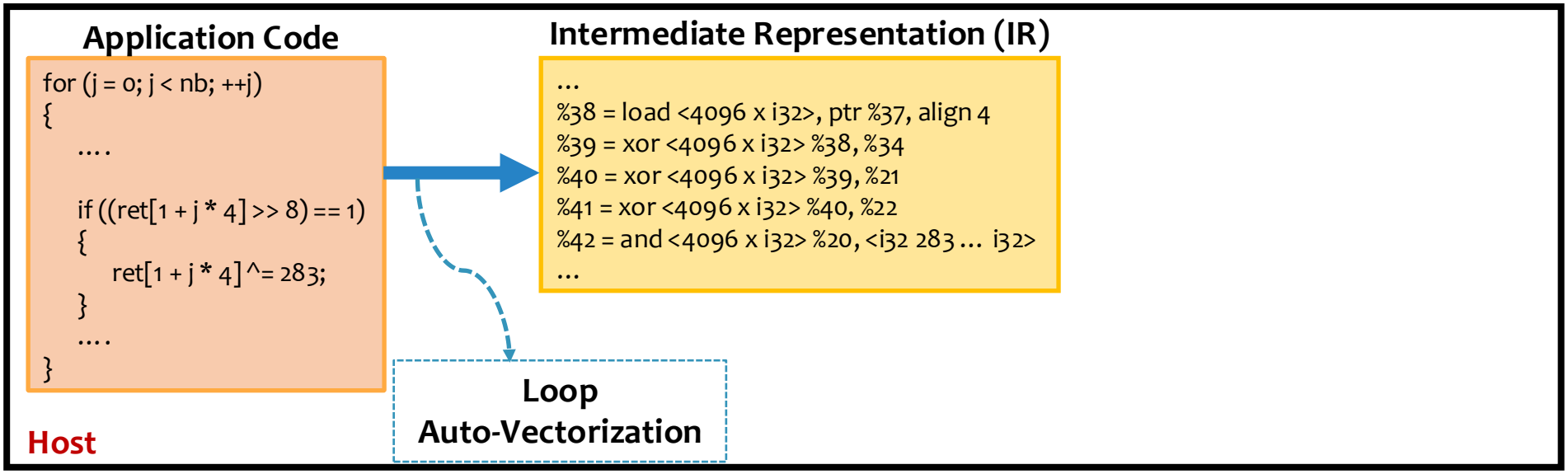
Runtime offloading of vectorized instructions to the most suitable SSD computation resource based on application and system characteristics

Conduit: Compile-Time Preprocessing (1/3)

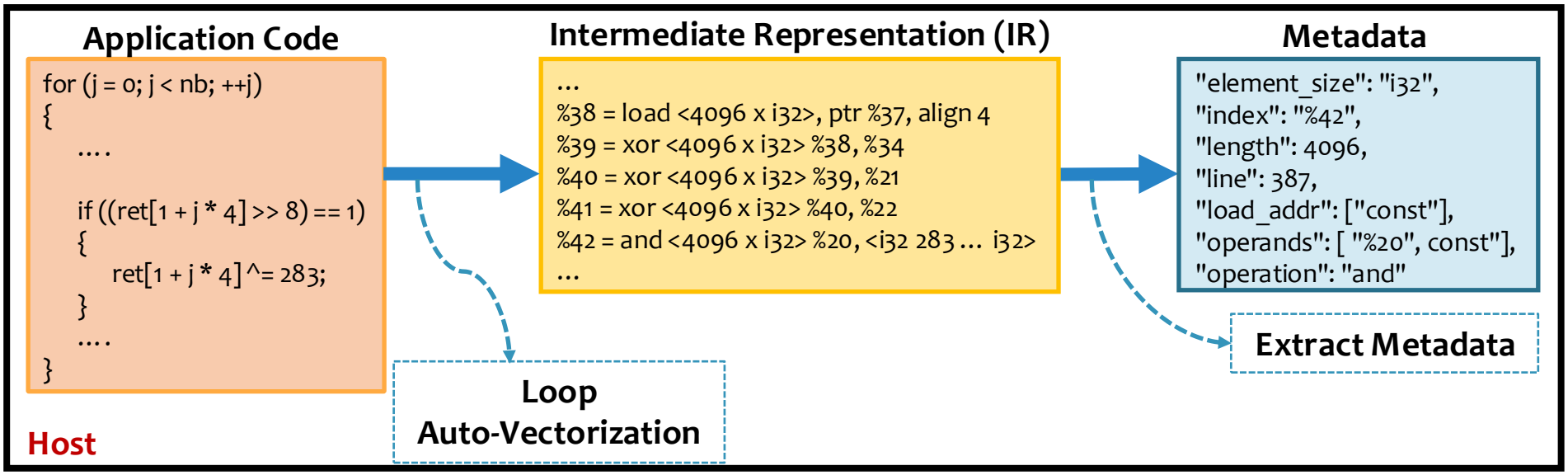


Conduit runs a custom compiler (e.g., LLVM) pass to identify offloadable code regions

Loop auto-vectorization transforms scalar instructions into wide vector operations that match the SSD-internal parallelism

Conduit leverages partial vectorization so that partially vectorizable code regions benefit from parallel execution in an SSD

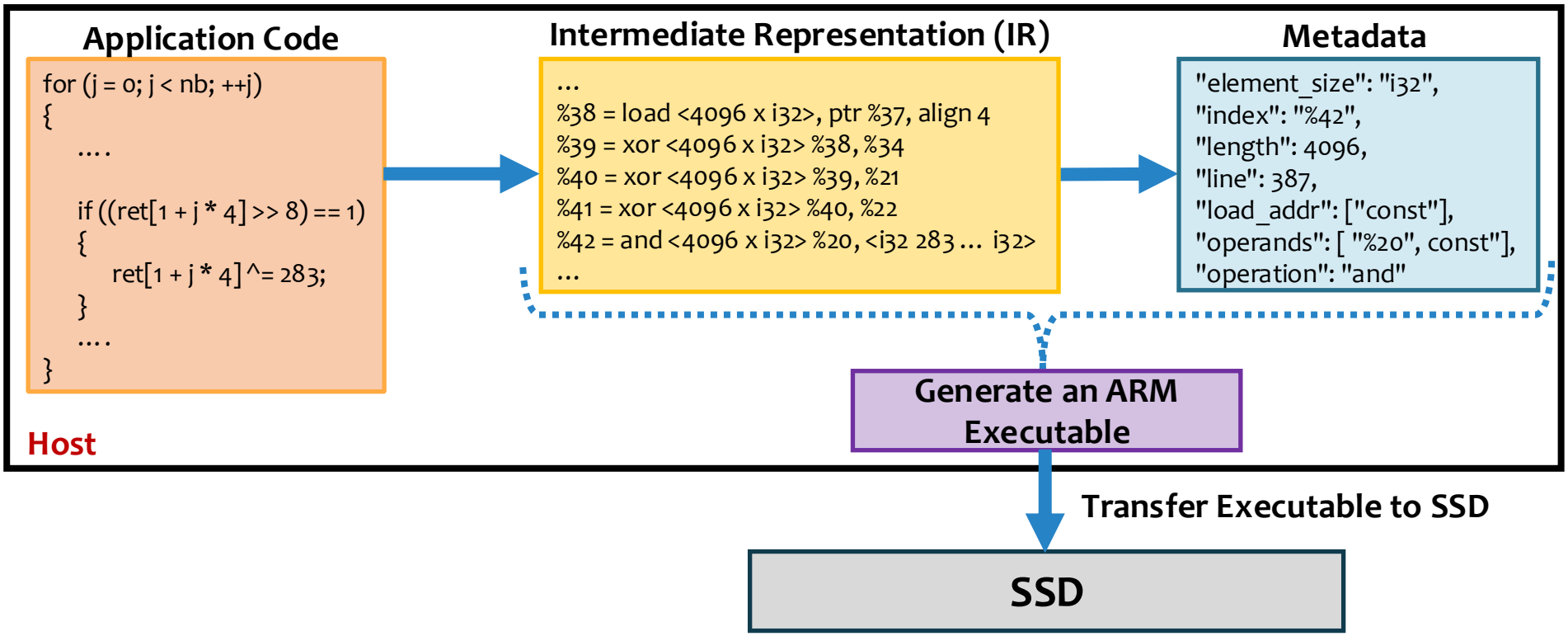
Conduit: Compile-Time Preprocessing (2/3)



Conduit extracts lightweight metadata (e.g., operation type, operand pointers, element sizes, vector length) for each vector operation

Conduit embeds the metadata with IR to reduce runtime scheduling latency and preserve programmer transparency

Conduit: Compile-Time Preprocessing (3/3)



Conduit converts the IR with metadata to an executable capable of running on ARM cores

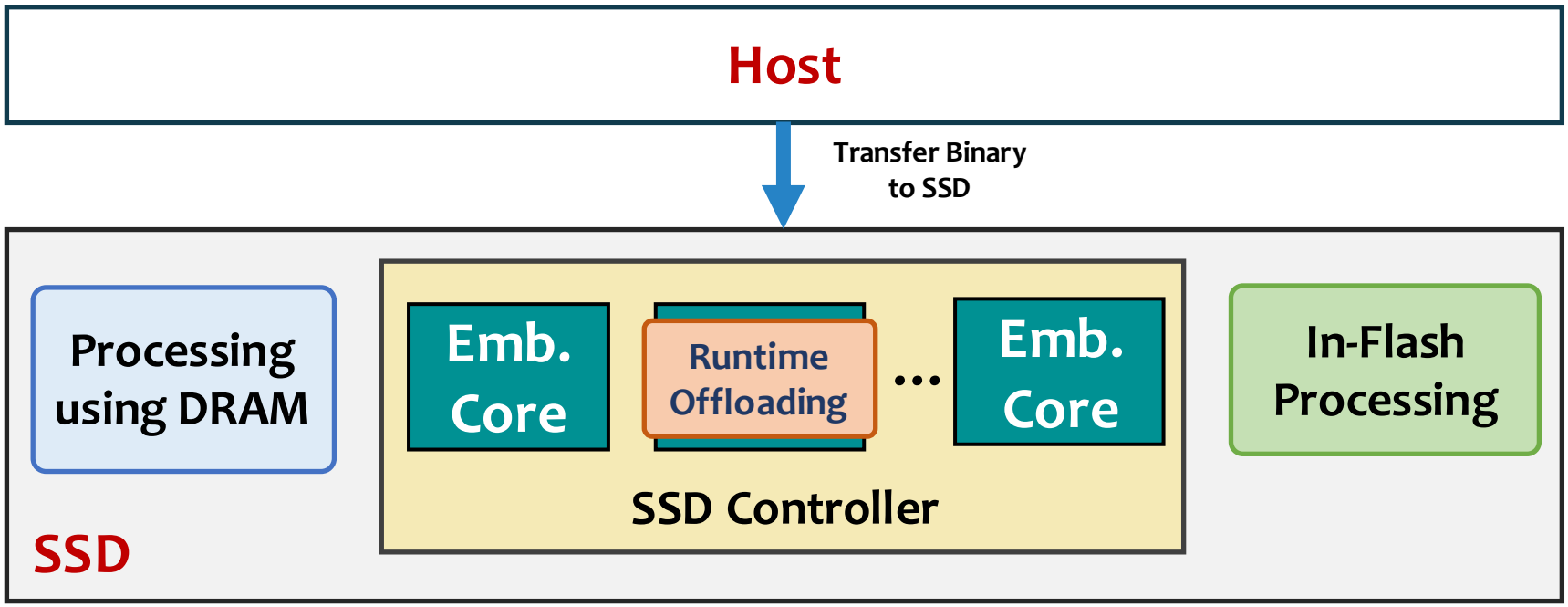
Conduit transfers the ARM executable to the SSD by repurposing existing NVMe admin commands for firmware update

Conduit: Key Steps

Compile-time pre-processing step to identify offloadable code regions in the application and transform them to wide vector operations

Runtime offloading of vectorized instructions to the most suitable SSD computation resource based on application and system characteristics

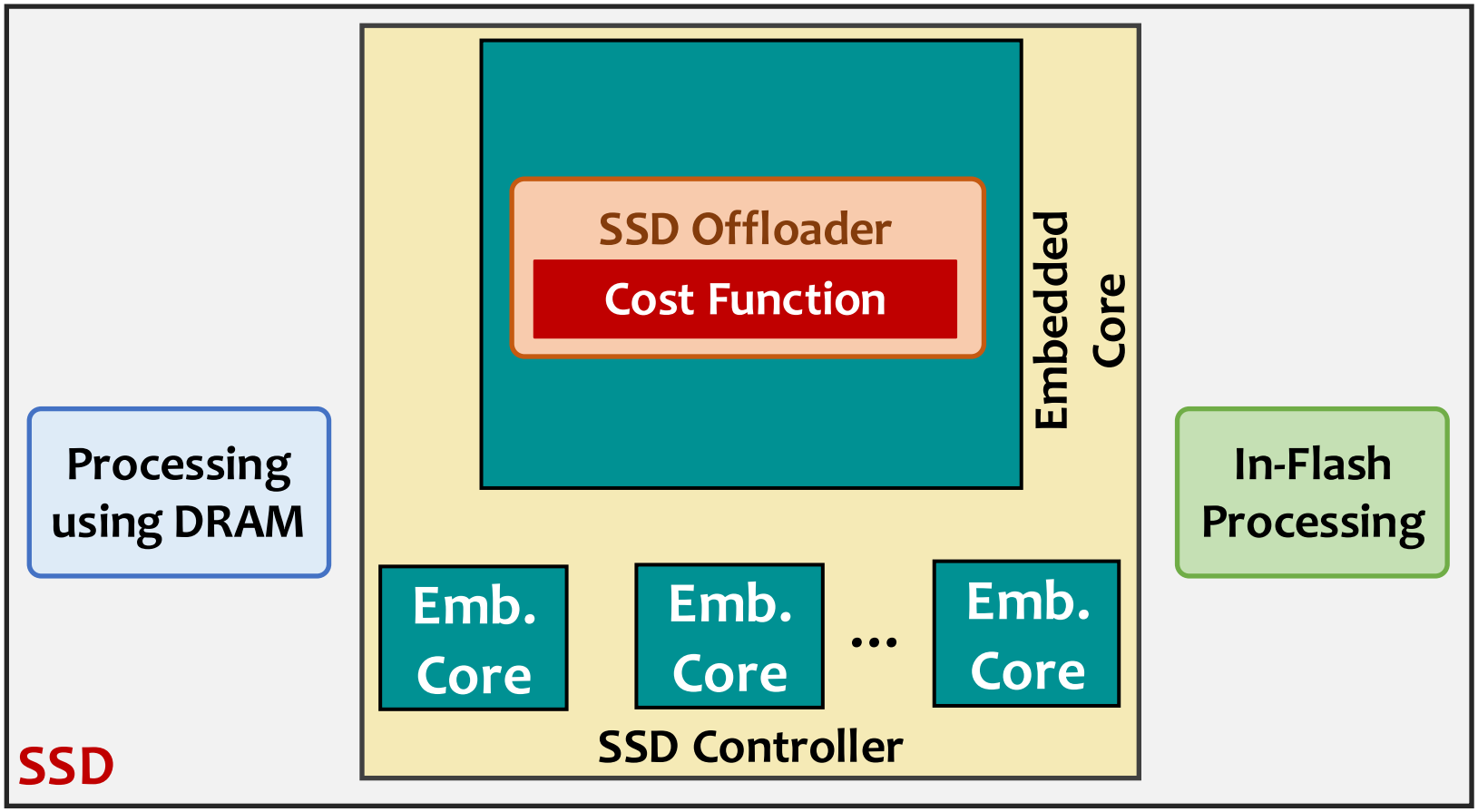
Conduit: Runtime Offloading (1/3)



Conduit performs runtime offloading in the SSD controller
to preserve programmer transparency and avoid changes to the storage stack

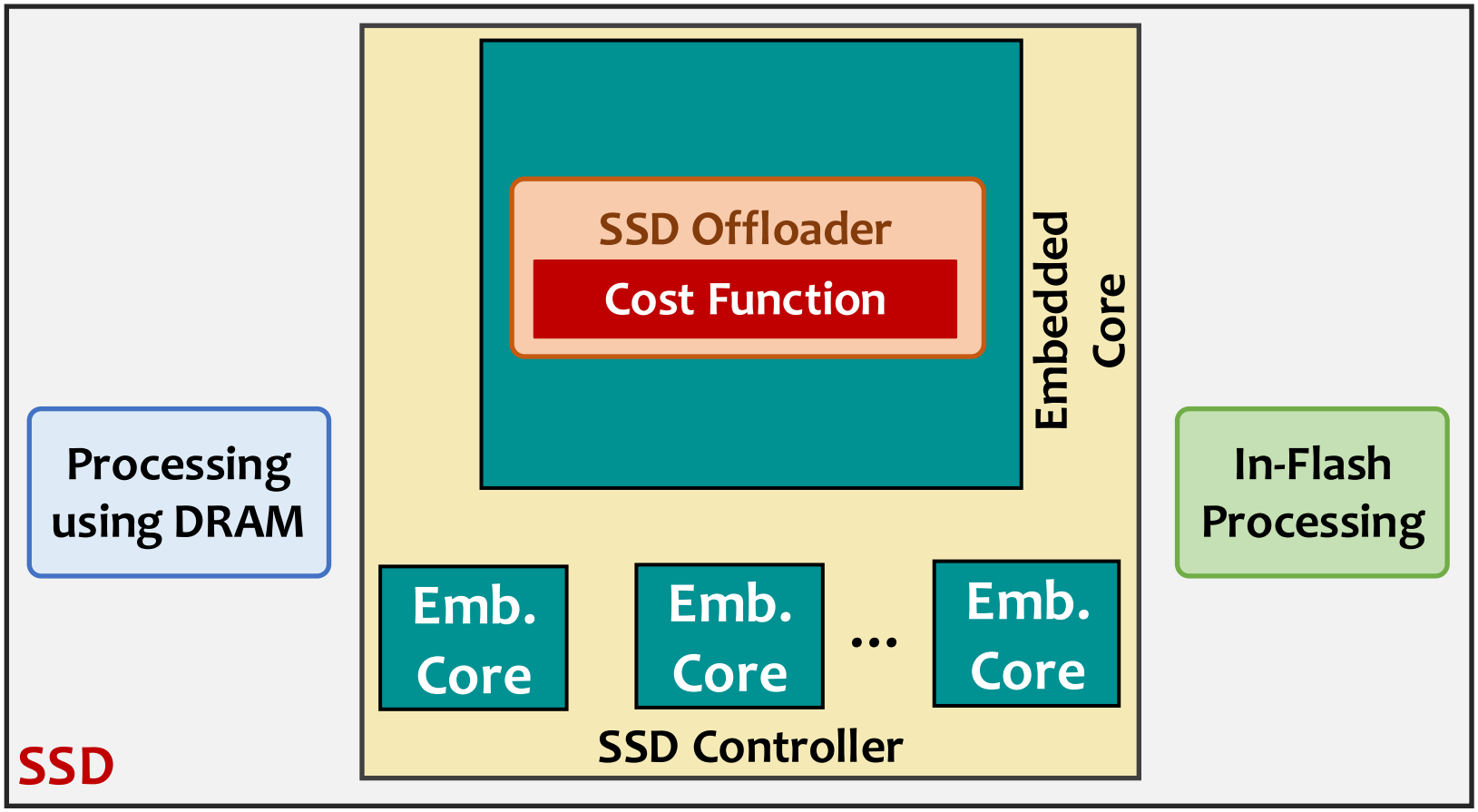
The SSD controller has real-time knowledge of the current state
(e.g., resource utilization, contention in shared channels)
of all SSD computation resources

Conduit: Runtime Offloading (2/3)



Conduit's SSD offloader maps each vectorized instruction in the ARM executable to the most-suitable SSD computation resource based on a cost function

Conduit: Runtime Offloading (3/3)



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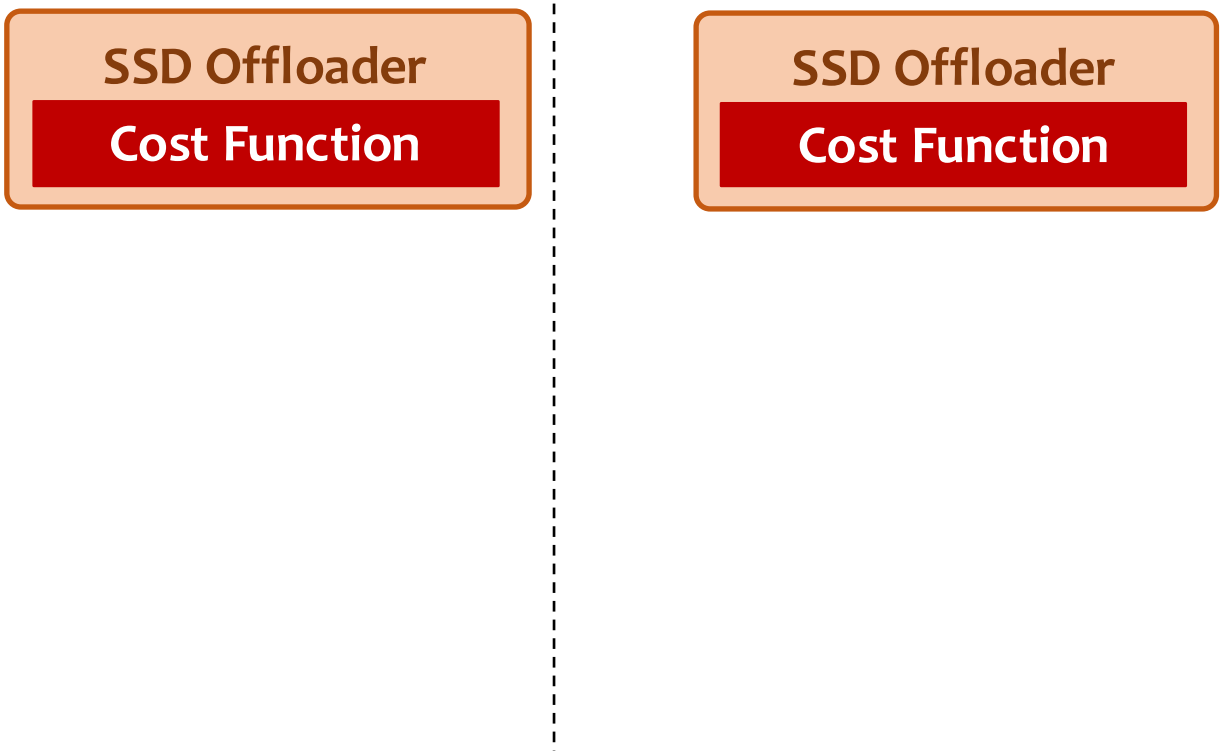
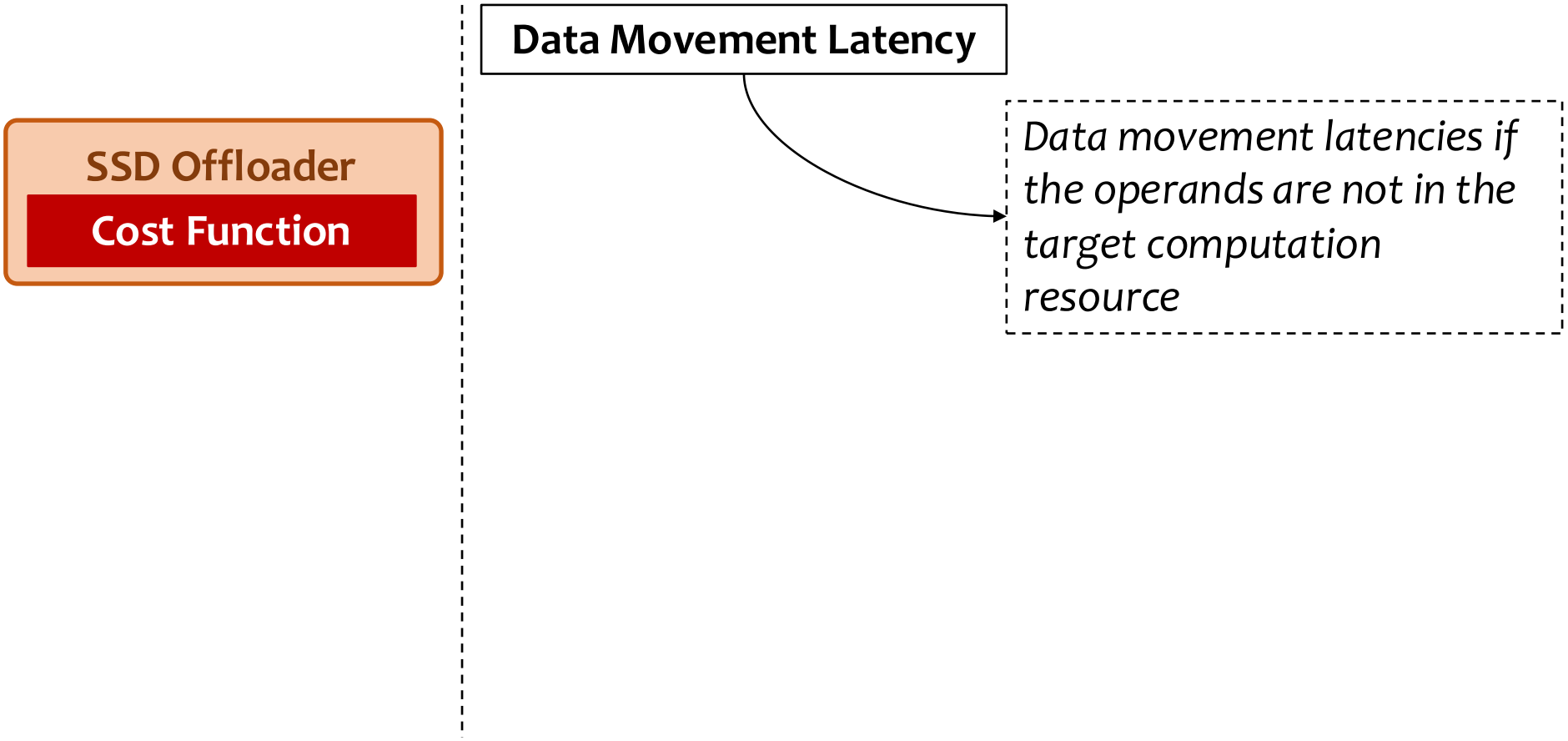
Conduit: Cost Function

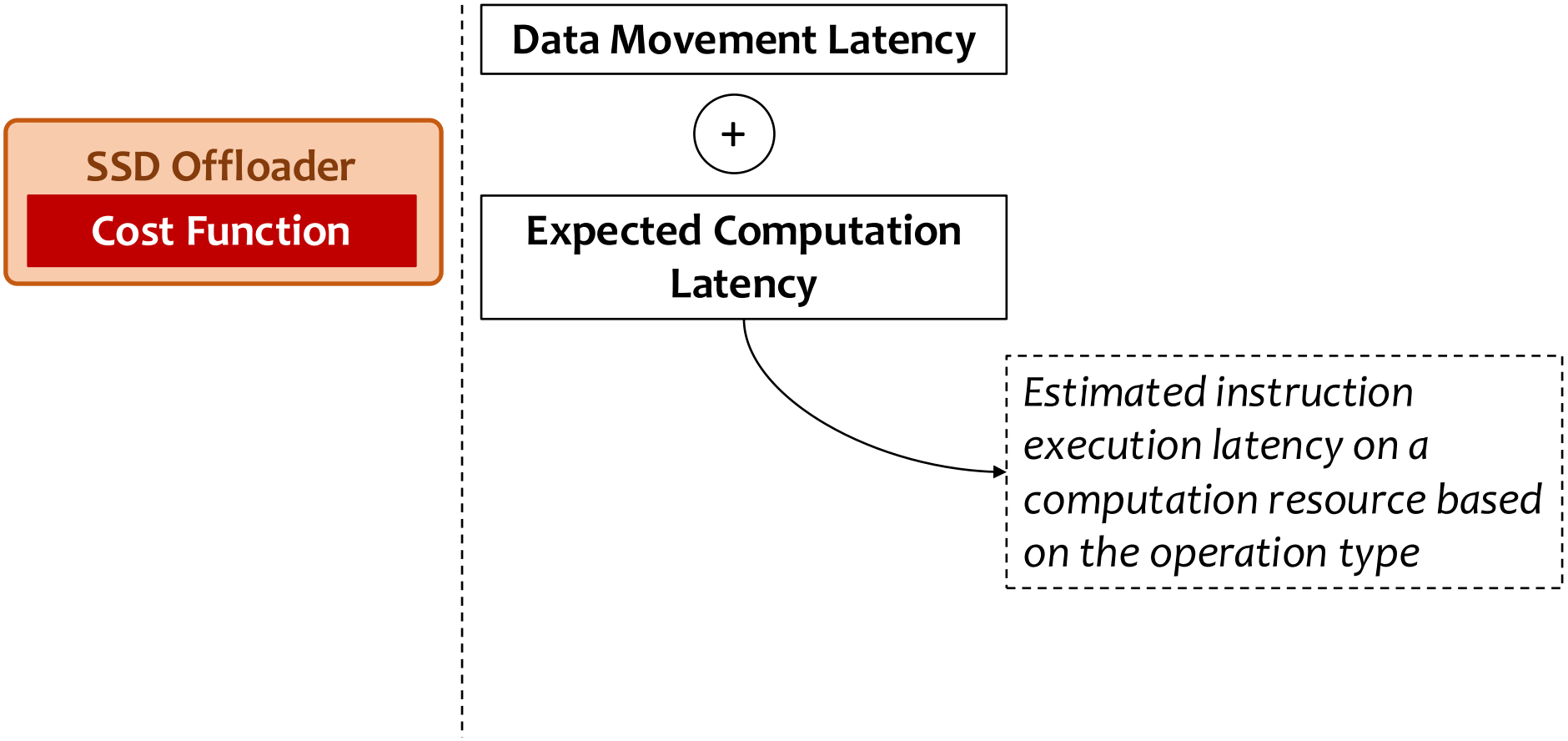


Conduit: Cost Function



Conduit tracks both application characteristics and current system conditions to evaluate the cost of offloading an instruction to each SSD computation resource

Conduit: Cost Function



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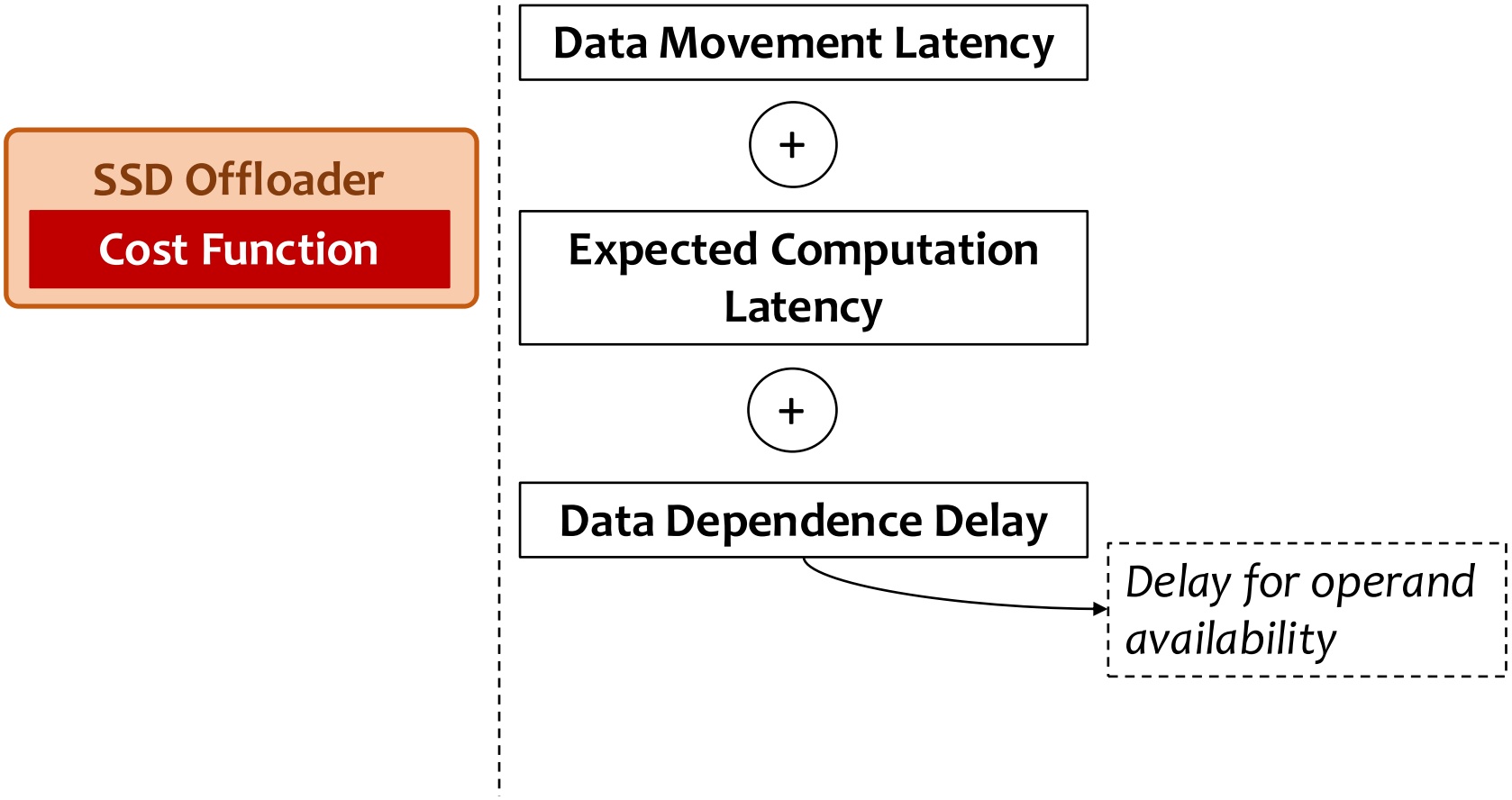

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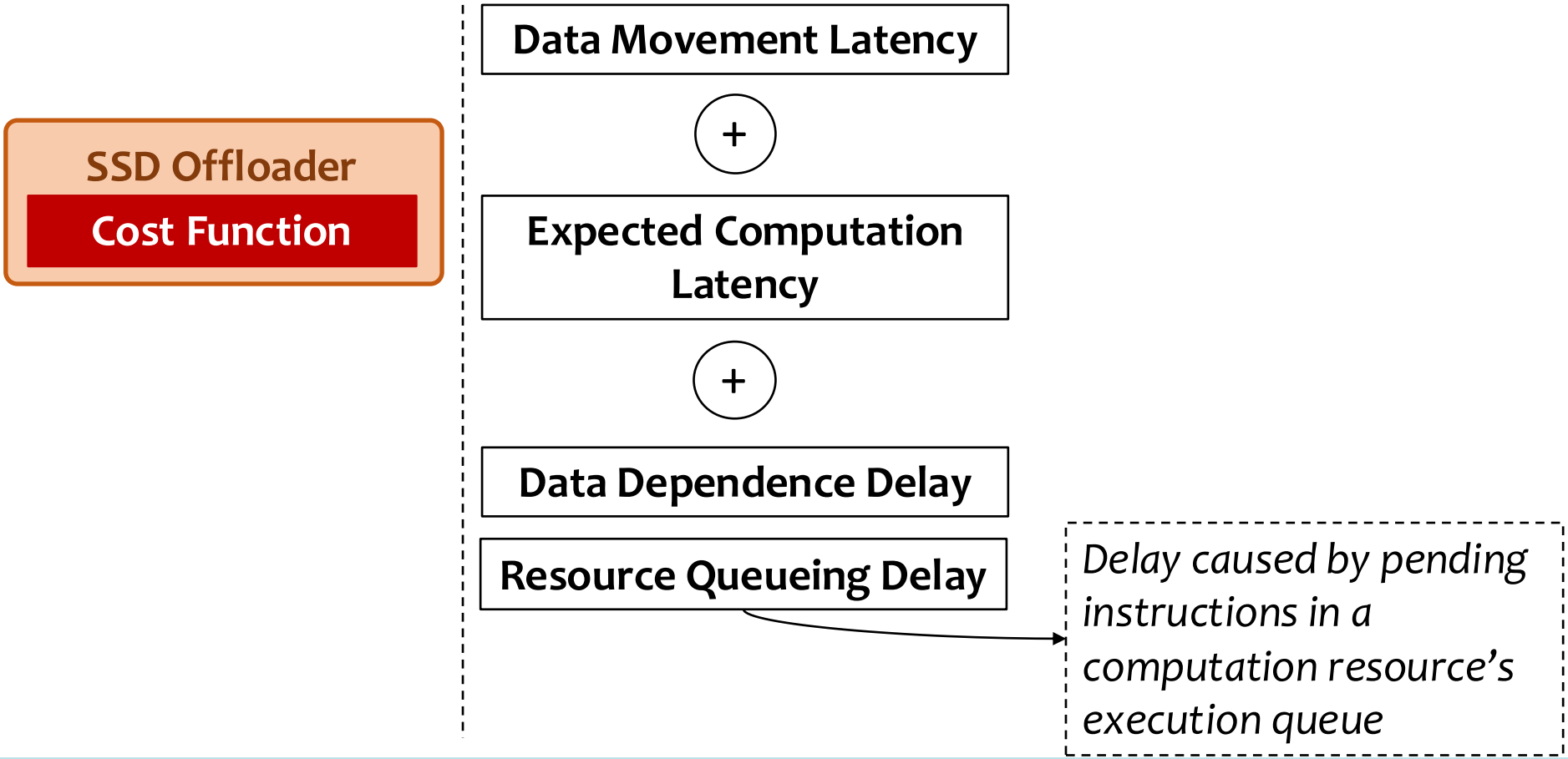

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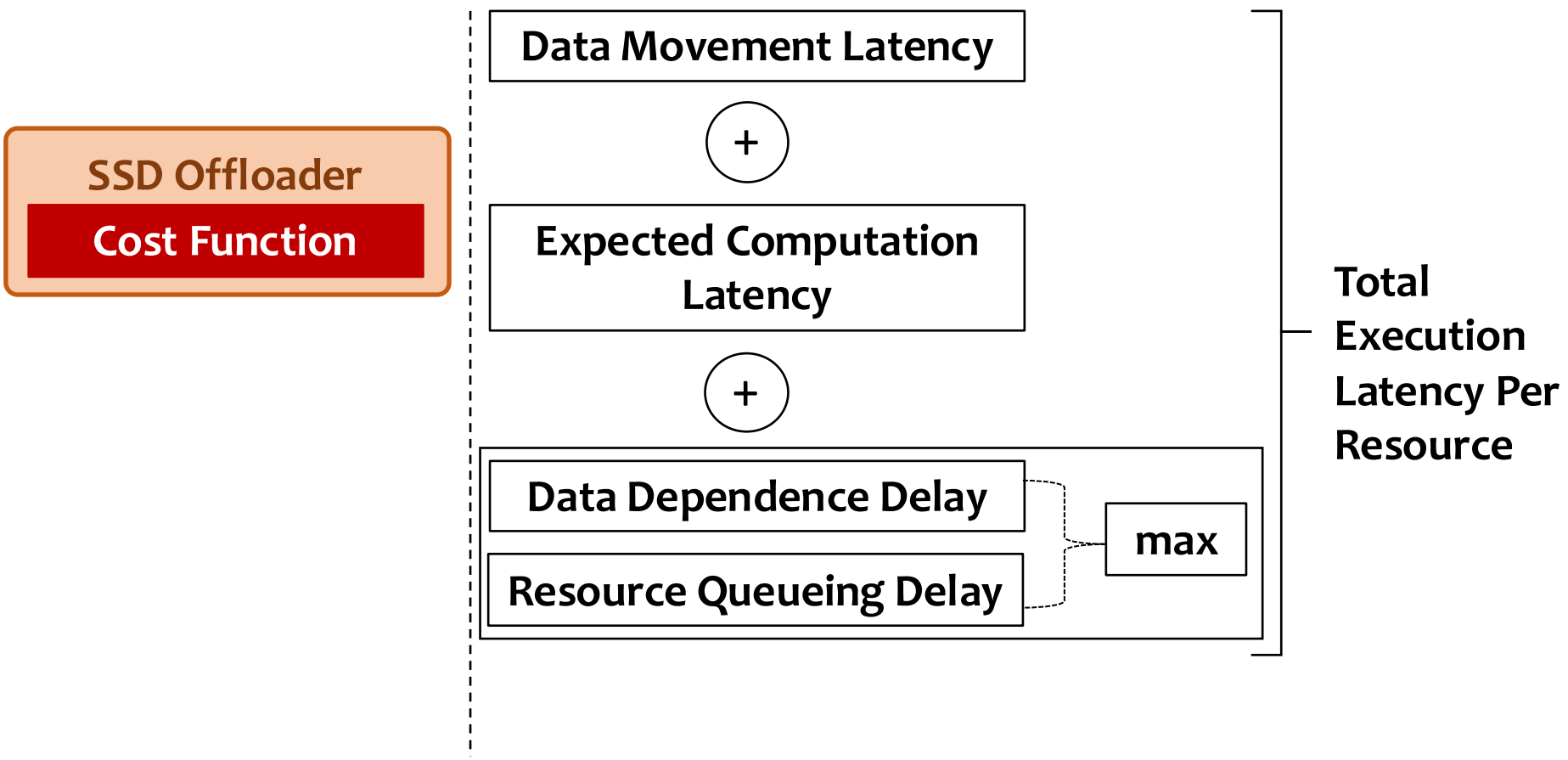

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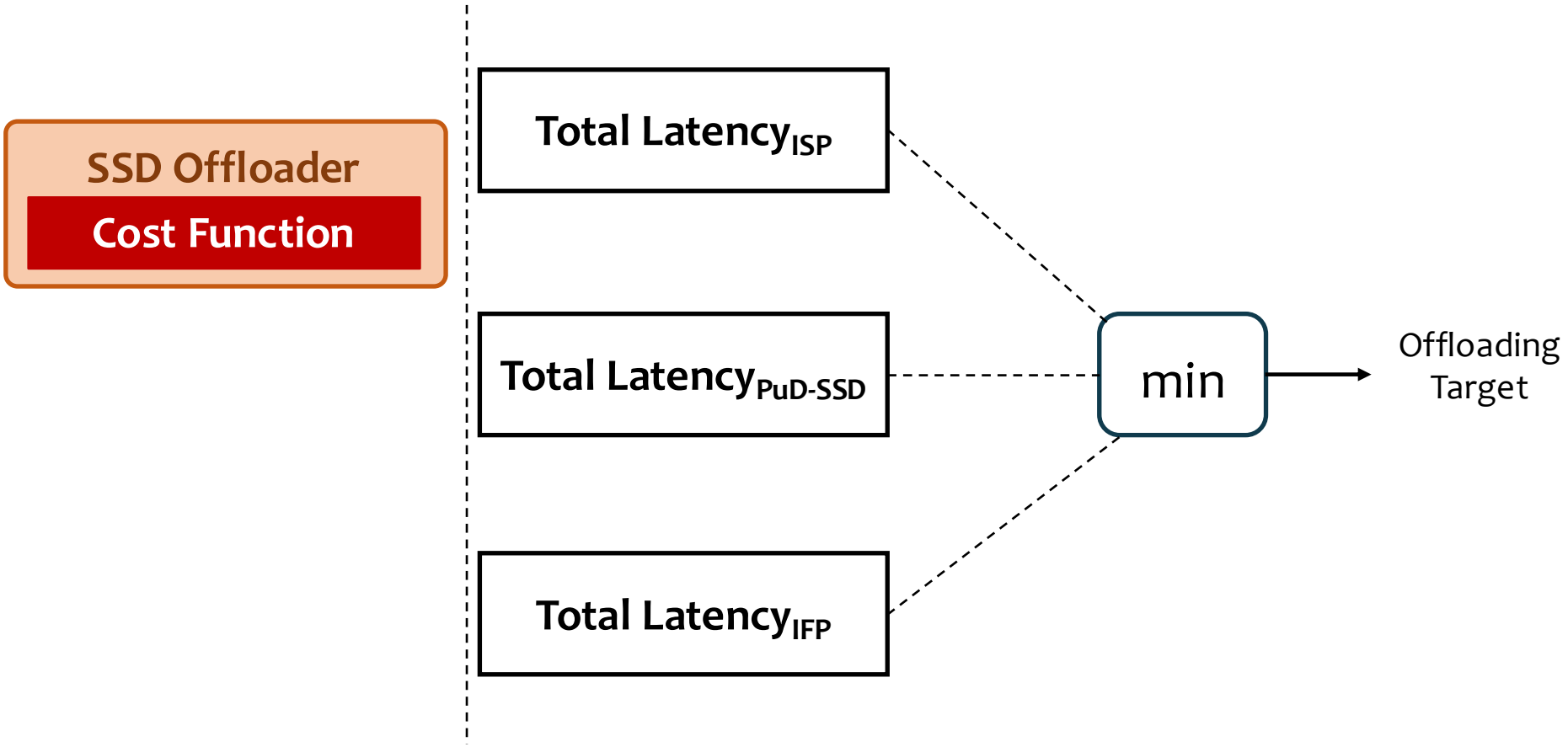
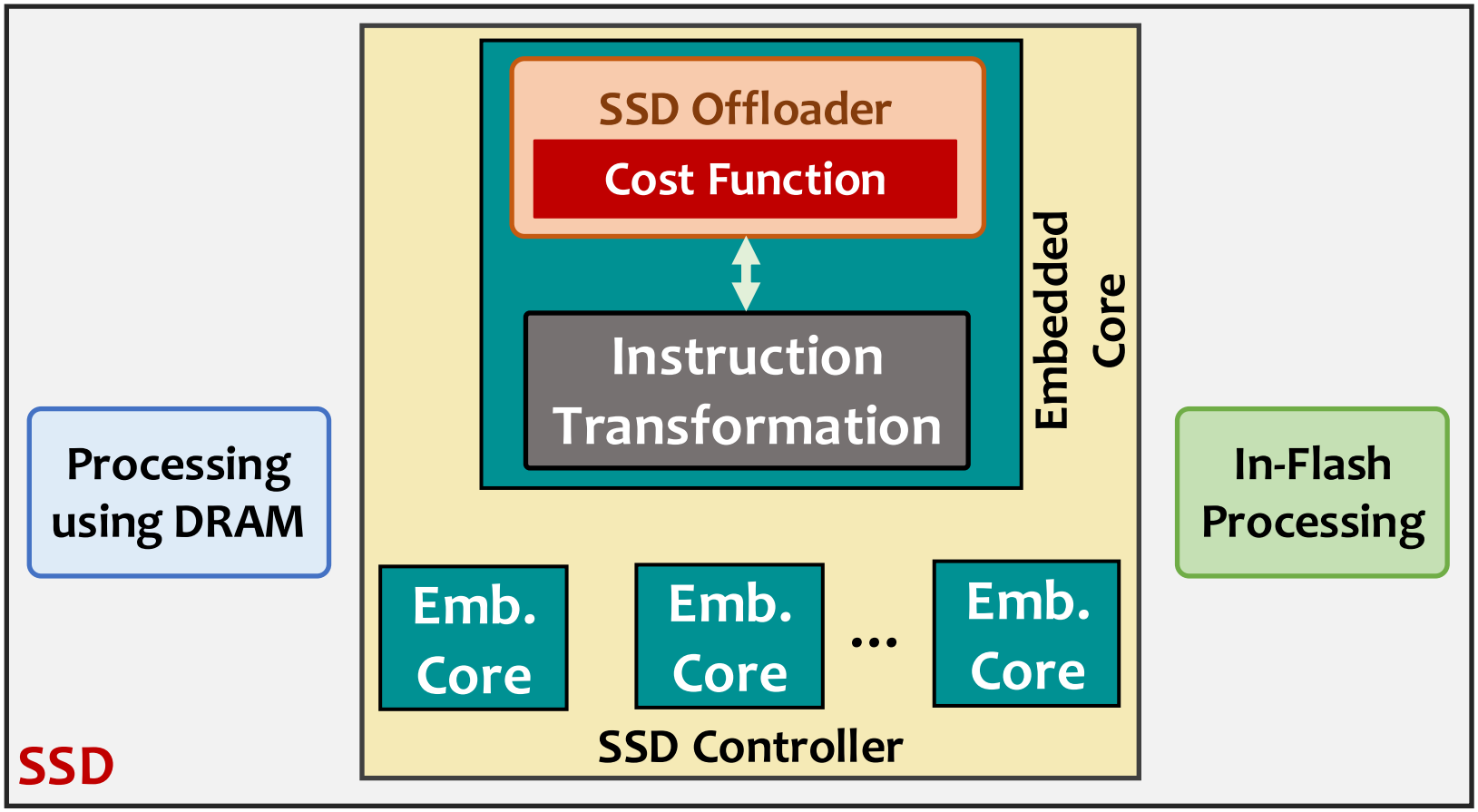


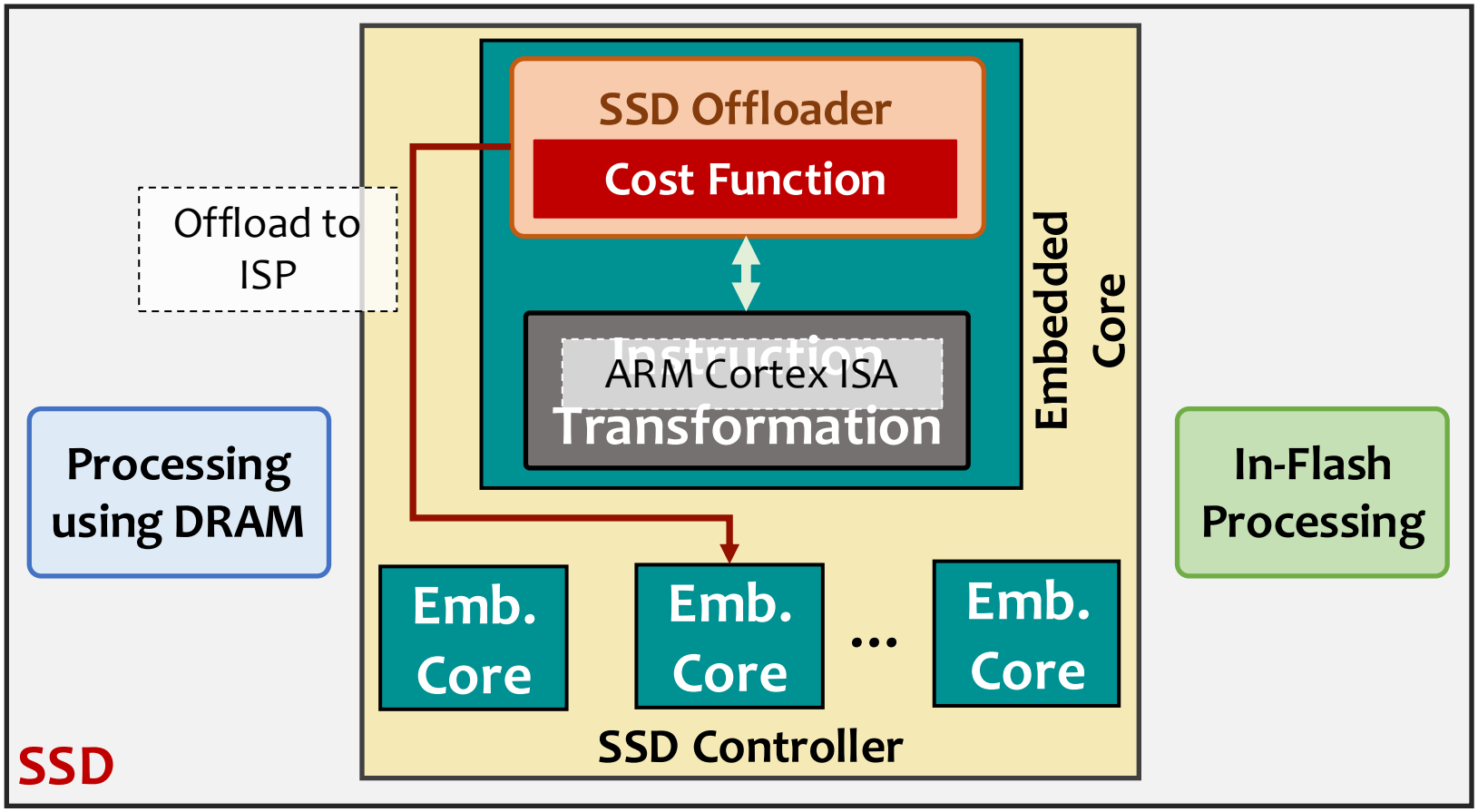
Conduit calculates the latency of executing each instruction on each computation resource, and selects the computation resource with the lowest execution latency

Conduit: Instruction Transformation (1/4)



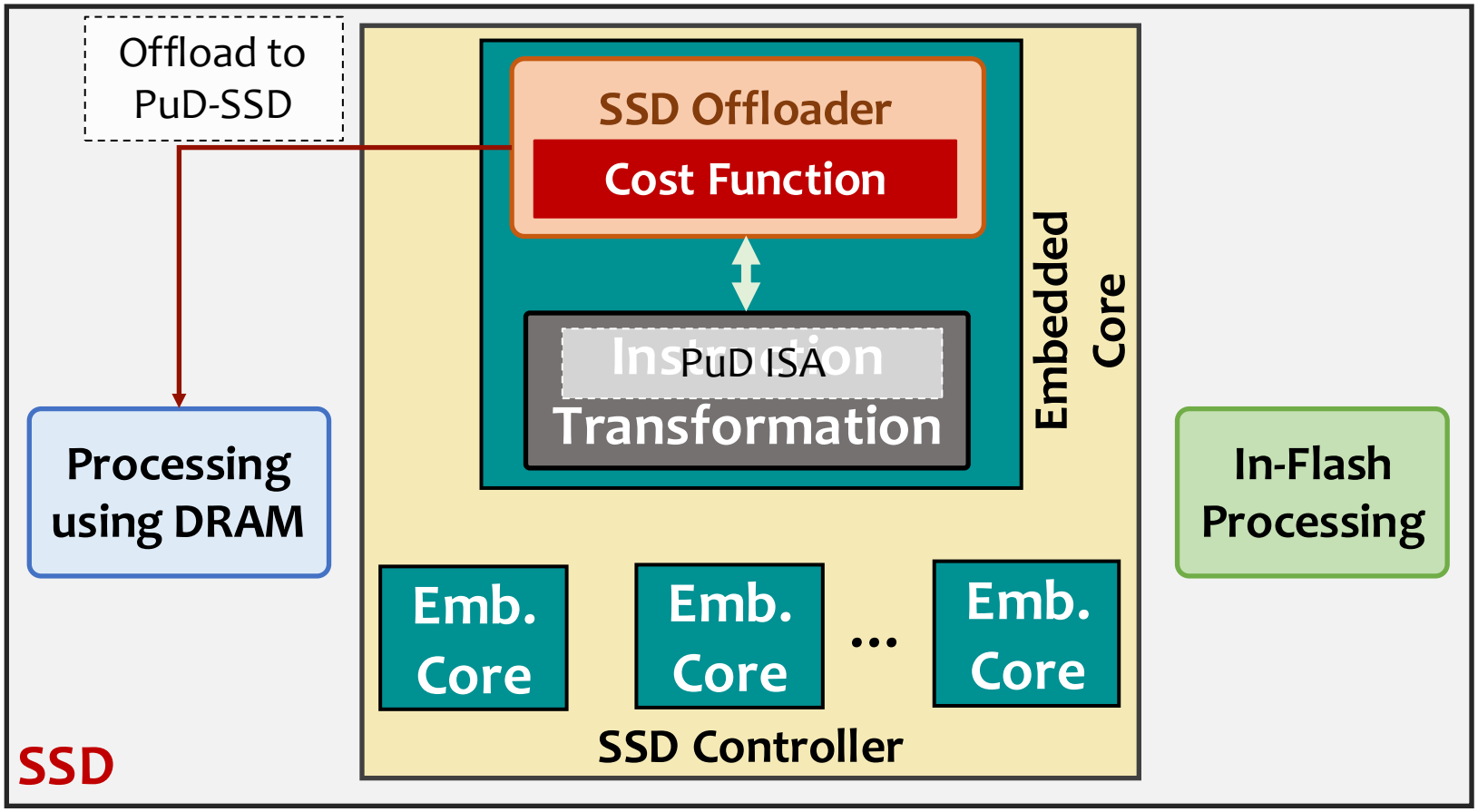
Conduit's instruction transformation unit transforms each vectorized instruction to the native ISA of the chosen SSD computation resource

Conduit: Instruction Transformation (2/4)



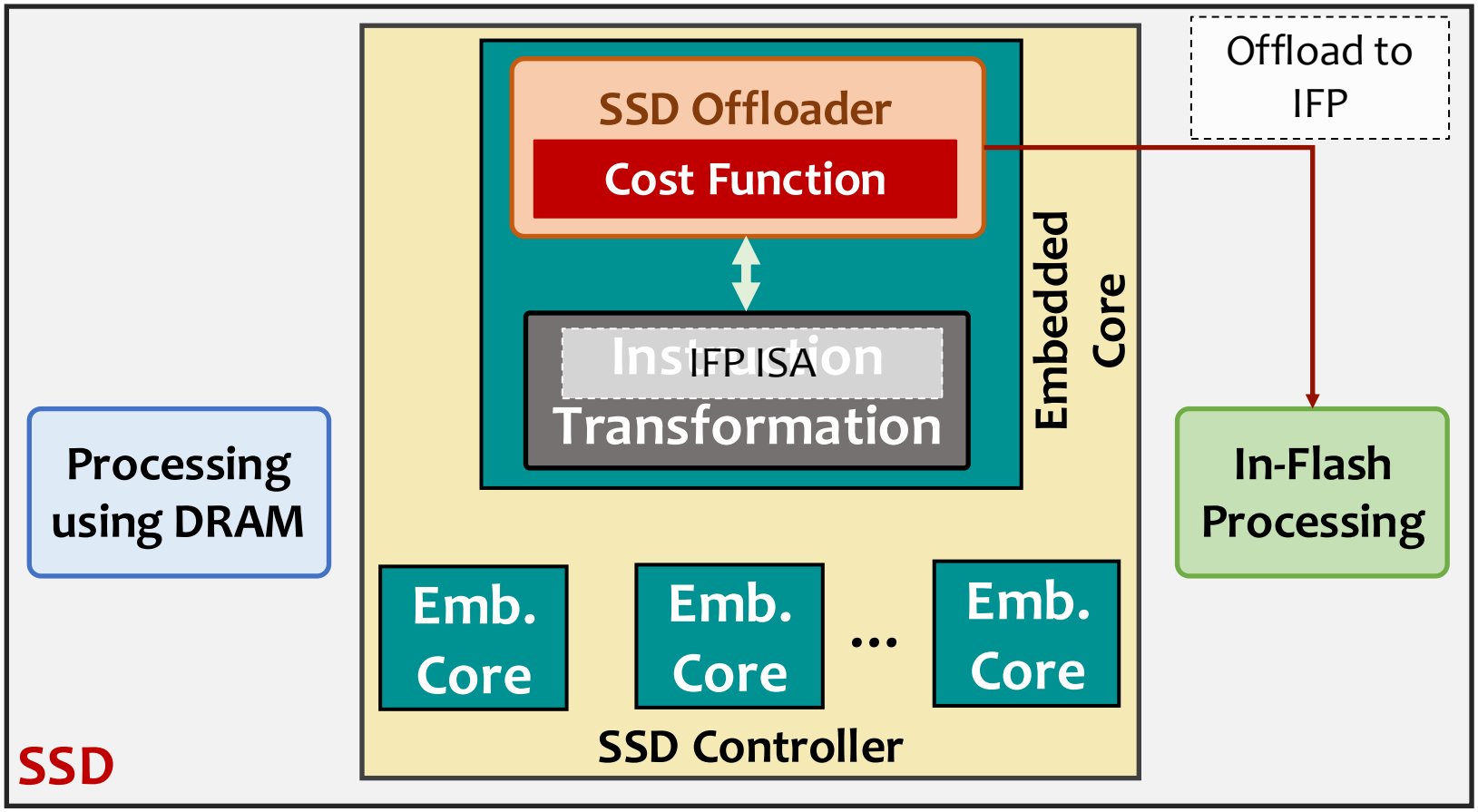
For ISP, Conduit transforms each vector operation to ARM Cortex ISA, and schedules the instruction for execution

Conduit: Instruction Transformation (3/4)



Conduit transforms each vector operation to ISA extensions (e.g., bbop_op for 2-input arithmetic) from prior PuD works (e.g., SIMDRAM [Hajinazar and Oliveira+, ASPLOS 2021])

Conduit: Instruction Transformation (4/4)



For IFP, Conduit leverages the MWS primitives from Flash-Cosmos [Park+, MICRO 2022] and arithmetic operations from Ares-Flash [Chen+, MICRO 2024] for instruction transformation

More in the paper

- Data coherence across SSD computation resources
  - Conduit employs a lazy coherence mechanism
- Data mapping and SSD-internal data movement
- Host-SSD communication
  - Regular I/O mode
  - Computation mode
- Failure handling
- Conduit's extensibility
- Limitations of auto-vectorization
- Conduit's applicability to irregular workloads
- Detailed background on the three NDP paradigms
- Detailed description of binary transfer from the host to the SSD

More in the Paper

Conduit: Programmer-Transparent Near-Data Processing Using Multiple Compute-Capable Resources in Solid State Drives

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Andreas Kosmas Kakolyris[†] Yu Liang[†‡] Mohammad Sadrosadati[†] Onur Mutlu[†]
[†]*ETH Zürich* [‡]*Inria, Paris*

https://arxiv.org/pdf/2601.17633

Talk Outline

Background and Motivation

Conduit

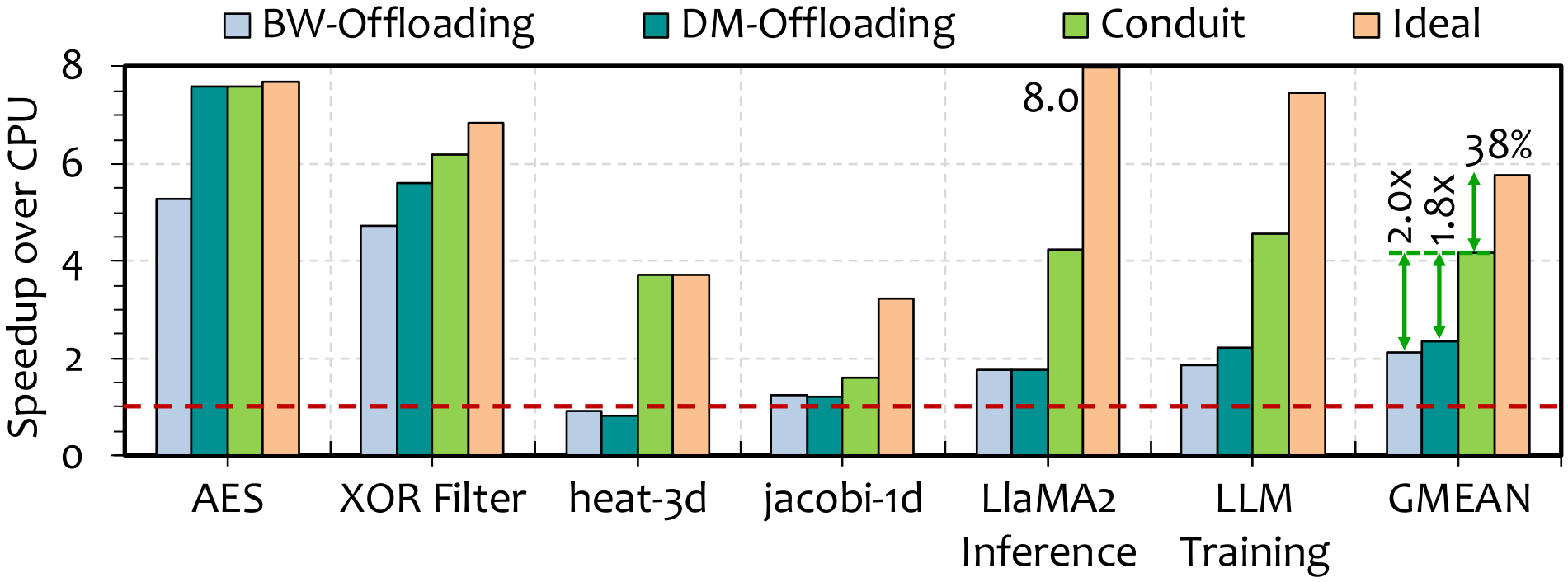
Evaluation

Summary

Evaluation Methodology

- Using an in-house event-driven SSD simulator that faithfully models:
  - Core SSD functionalities (e.g., address translation, transaction scheduling) based on MQSim [Tavakkol+, FAST 2018]
  - DRAM modeling based on Ramulator [Kim+, CAL 2015]
  - Shared resource contention
  - Performance models of three NDP paradigms
- SSD configuration based on Samsung 980 Pro SSD
- Six data-intensive workloads from
  - AES, XOR Filter, heat-3d, jacobi-1d, LlaMA2 training and inference
- Evaluated Policies
  - Host CPU
  - BW-Offloading: Offloading based on bandwidth utilization (based on TOM [Hsieh+, ISCA 2016], ƛ-IO [Yang+, FAST 2023])
  - DM-Offloading: Offloading based on data movement cost (based on ALP [Mansouri Ghiasi+, TETC 2023], PIMProf [Wei+, DATE 2022])
  - Conduit
  - Ideal Offloading: No resource contention, resource selection based on lowest computation latency and zero data movement

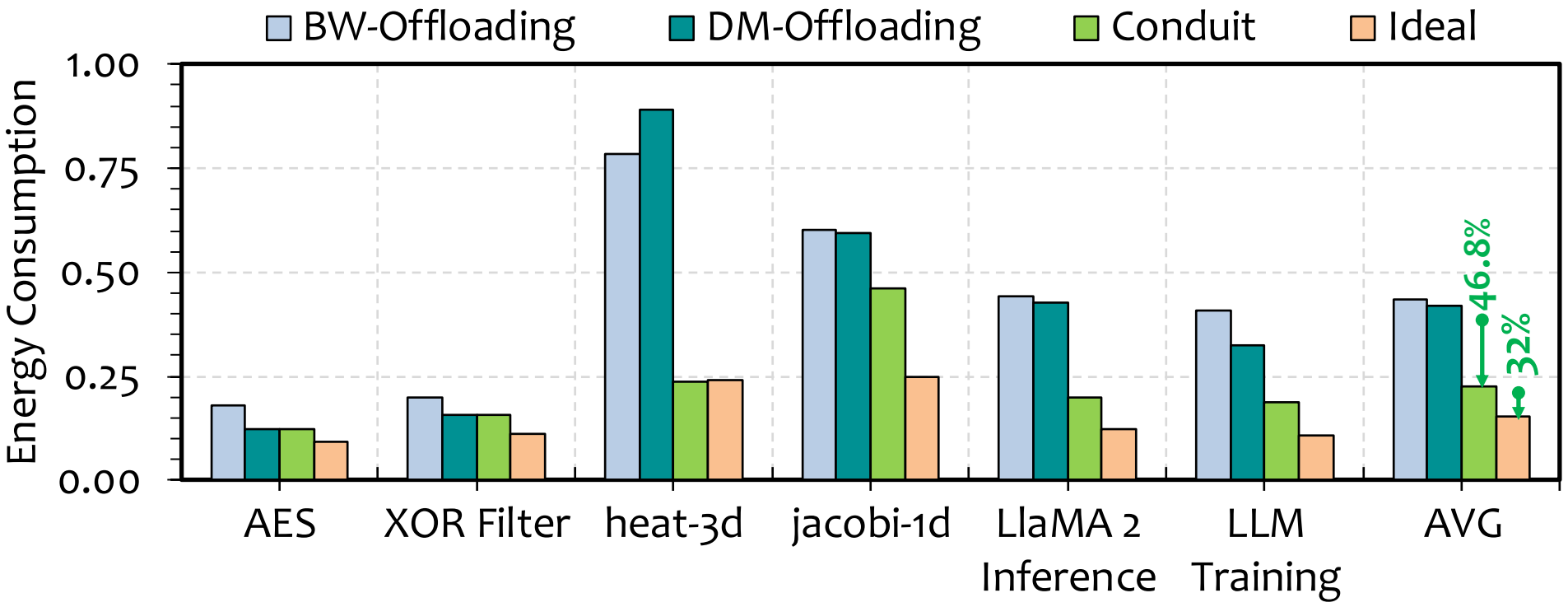
Results: Performance Analysis



Conduit outperforms the best-performing prior approach by 1.8x on average by making holistic offloading decisions

Conduit provides 62% of the Ideal offloading policy's performance without the ideal approach's unrealistic assumptions

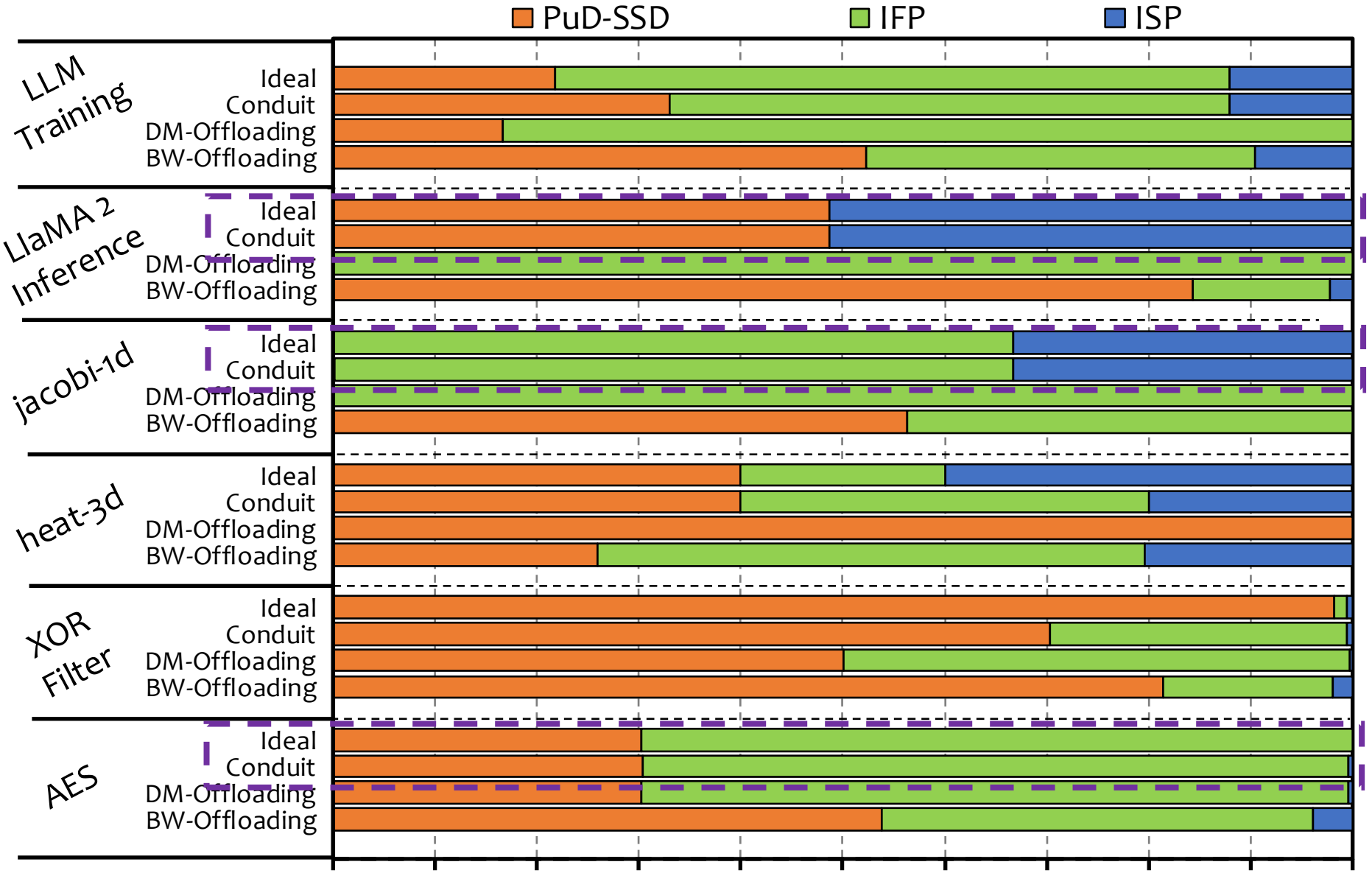
Results: Energy Consumption



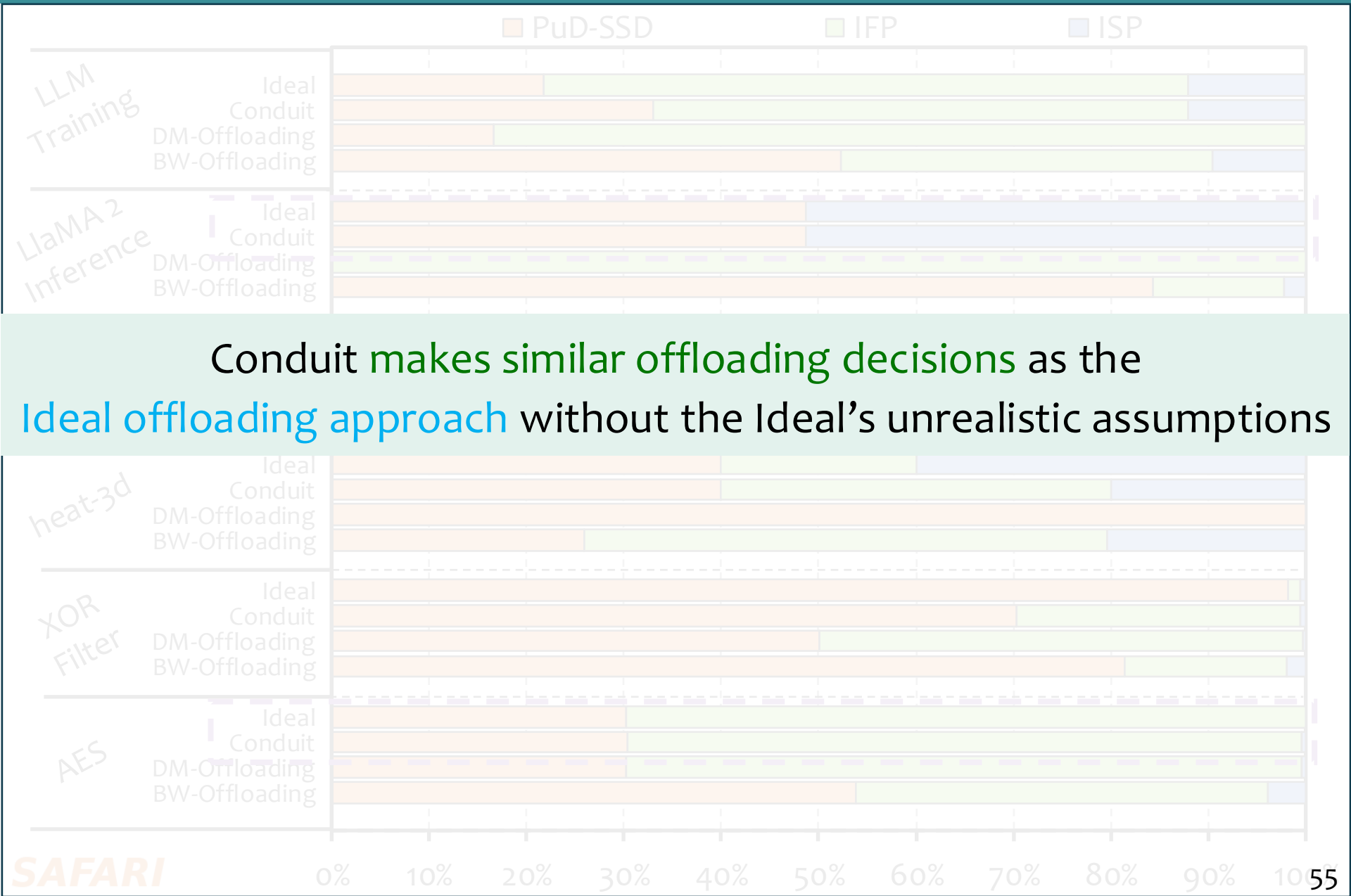
Conduit reduces energy consumption by 46.8% on average compared to DM-Offloading by exploiting all SSD computation resources and reducing resource queueing delays

Conduit provides 68% of the energy efficiency of the Ideal approach

Offloading Decisions



Offloading Decisions



Conduit makes similar offloading decisions as the Ideal offloading approach without the Ideal's unrealistic assumptions

Overhead Analysis

- **Storage Overhead**

17 bytes for storing cost function features in the SSD DRAM

1.5 KiB for storing the instruction transformation mapping of vectorized instructions to the native ISA of all computation resources

- **Runtime Latency Overhead**

3.8 µs on average for cost function feature collection and instruction transformation for each instruction

- **Area Overhead**

Conduit does not add any area overhead beyond that of the prior techniques it leverages

More in the Paper

- Tail latency evaluation
  - In LlaMA2 inference, compared to DM-Offloading, Conduit reduces 99th percentile latency by 5.6x, and 99.99th percentile latency by 22.3x
- Interaction between the workload and SSD computation resources
- Detailed description of overhead analysis
- Detailed workload description and characteristics
- Detailed description of SSD modeling and NDP extensions

More in the Paper

Conduit: Programmer-Transparent Near-Data Processing Using Multiple Compute-Capable Resources in Solid State Drives

Rakesh Nadig[†] Vamanan Arulchelvan[†] Mayank Kabra[†] Harshita Gupta[†]
Rahul Bera[†] Nika Mansouri Ghiasi[†] Nanditha Rao[†] Qingcai Jiang[†]
Andreas Kosmas Kakolyris[†] Yu Liang[†‡] Mohammad Sadrosadati[†] Onur Mutlu[†]
[†]*ETH Zürich* [‡]*Inria, Paris*

https://arxiv.org/pdf/2601.17633

Talk Outline

Background and Motivation

Conduit

Evaluation

Summary

Conduit: Summary

First general-purpose programmer-transparent NDP framework that enables fine-grained computation offloading across multiple SSD computation resources

Improves performance by 1.8x on average over the best-performing prior offloading approach

Reduces energy consumption by 46.8% on average over the most efficient prior work

Low latency and storage overheads

Conduit

Programmer-Transparent Near-Data Processing Using Multiple Compute-Capable Resources in Solid State Drives

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HPCA 2026

SAFARI Research Group
safari.ethz.ch

ETH zürich

Inria

Programmer-Transparent Near-Data Processing Using Multiple Compute-Capable Resources in Solid-State Drives

Rakesh Nadig

28 June 2026

6th Workshop on Memory-Centric Computing Systems

ISCA 2026

SAFARI

ETH zürich

FAQ (1/2)

- Executive Summary
- PuD-SSD
- IFP
- Extended Case Study (With More Compute-Intensive Workloads)
- Compile-Time Preprocessing
- Runtime Offloading
- System Integration

FAQ (2/2)

- Evaluated Configurations
- Workload Characteristics
- Results
  - Extended Performance Results
  - Extended Energy Consumption
  - Tail Latency
  - Workload-Computation Resource Interaction

Executive Summary

- **Background:** A solid-state drive enables three near-data processing paradigms: processing using embedded cores, processing using DRAM, and in-flash processing
- **Problem:**
  - Prior SSD-based NDP techniques (1) do not exploit the SSD's full computational potential, (2) are application-specific, and (3) are not programmer-transparent
  - Prior main-memory-based NDP offloading techniques provide limited benefits for an SSD because they optimize for limited factors (e.g., bandwidth utilization)
- **Goal:** To enable a programmer-transparent NDP framework in SSDs that (1) offloads computations across multiple SSD computation resources, and (2) makes offloading decisions that are both workload- and system-aware
- **Conduit: A** general-purpose programmer-transparent NDP framework that consists of:
  - compile-time preprocessing to identify offloadable code regions and transform them to wide vector operations,
  - runtime offloading that maps computations to the most suitable SSD computation resource based on workload and system factors
- **Key Results:** We evaluate Conduit and two prior NDP offloading techniques using an in-house event-driven SSD simulator on six data-intensive workloads
  - Compared to the best prior offloading policy, Conduit improves performance by 1.8x on average and reduces energy consumption by 46% on average

In-Storage Processing (ISP)

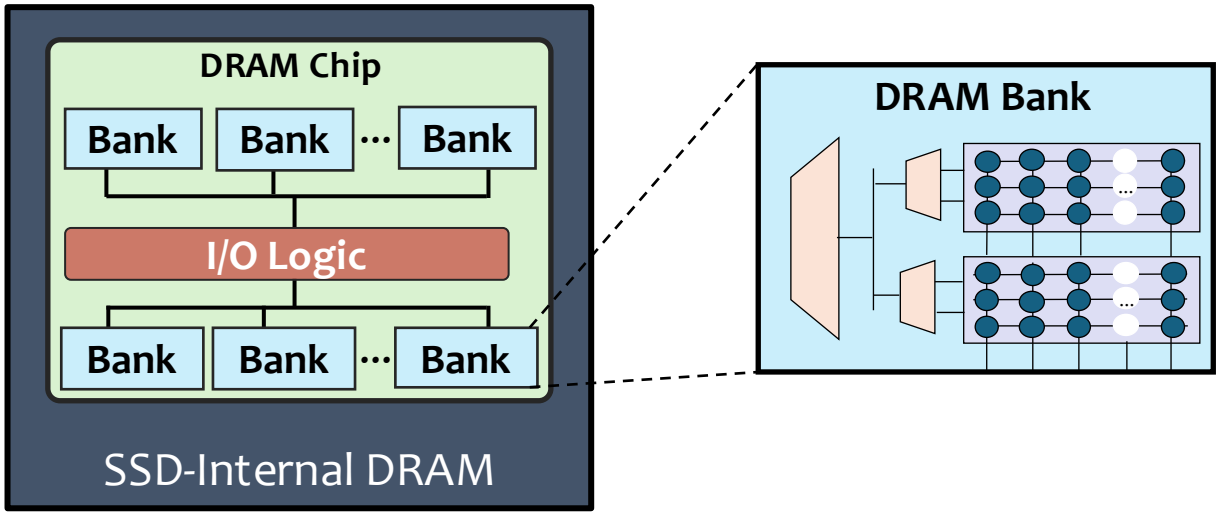
The embedded cores in the SSD controller typically execute FTL firmware and handle host communication

The embedded cores can be repurposed to perform general-purpose computations

Many prior works show that ISP accelerates various tasks including filtering, aggregation, encryption, and search

Limited SIMD parallelism (e.g., 32-bit registers in ARM Cortex R5)

Processing-Using-DRAM in SSD (PuD-SSD)



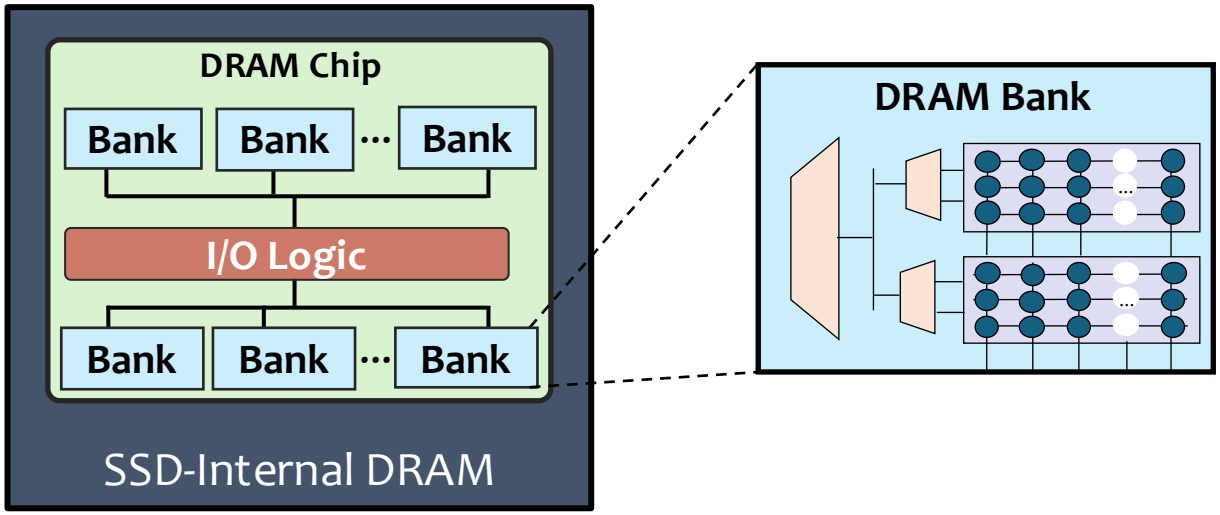
PuD techniques leverage DRAM's operational principles to perform computations within the memory array

Prior works (e.g., SIMDRAM[1], MIMDRAM[2]) enable bulk-bitwise and complex arithmetic operations entirely within the DRAM chips

[1] Hajinazar+, "SIMDRAM: A Framework for Bit-Serial SIMD Processing using DRAM," In ASPLOS, 2021

[2] Oliveira+, "MIMDRAM: An End-to-End Processing-Using-DRAM System for High-Throughput, Energy-Efficient and Programmer-Transparent Multiple-Instruction Multiple-Data Computing," In HPCA, 2024

Processing-Using-DRAM in SSD (PuD-SSD)



PuD-SSD requires transferring operands from flash chips to the DRAM via bandwidth-limited flash channels

These data transfers incur latency and energy overheads, and cause contention in flash channels

[1] Hajinazar+, "SIMDRAM: A Framework for Bit-Serial SIMD Processing using DRAM," In ASPLOS, 2021

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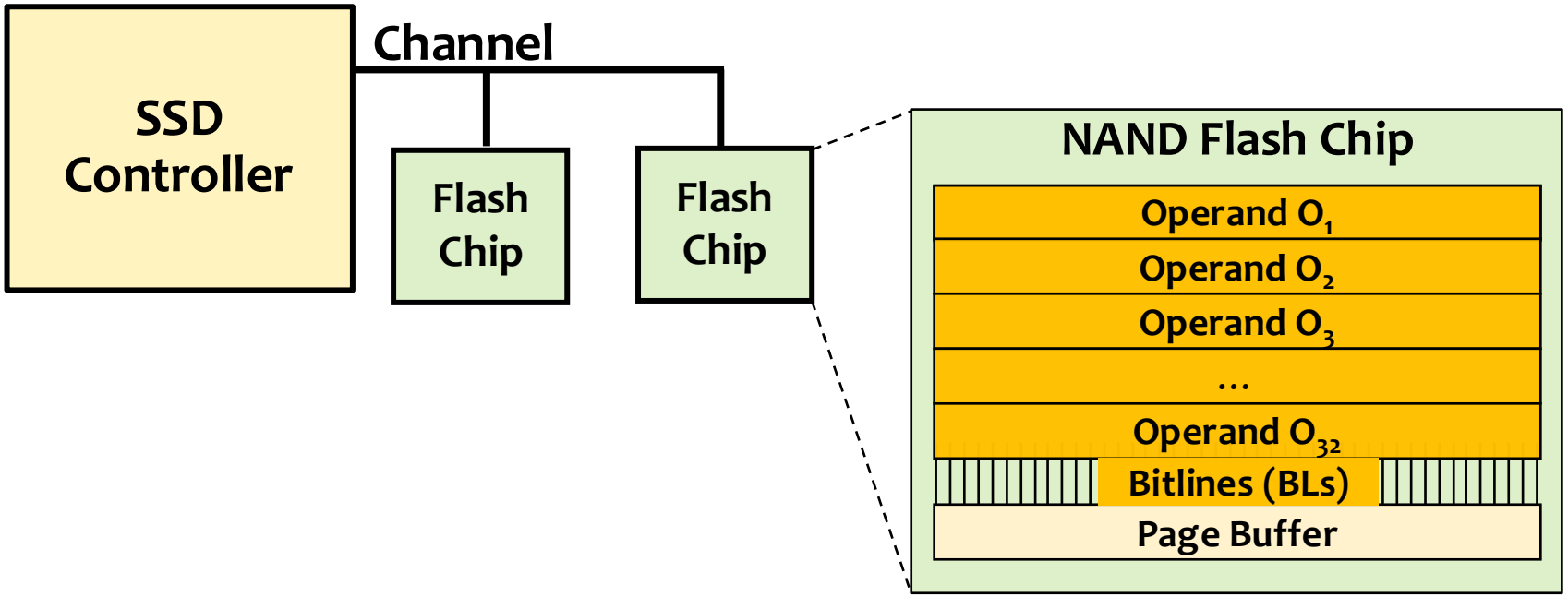
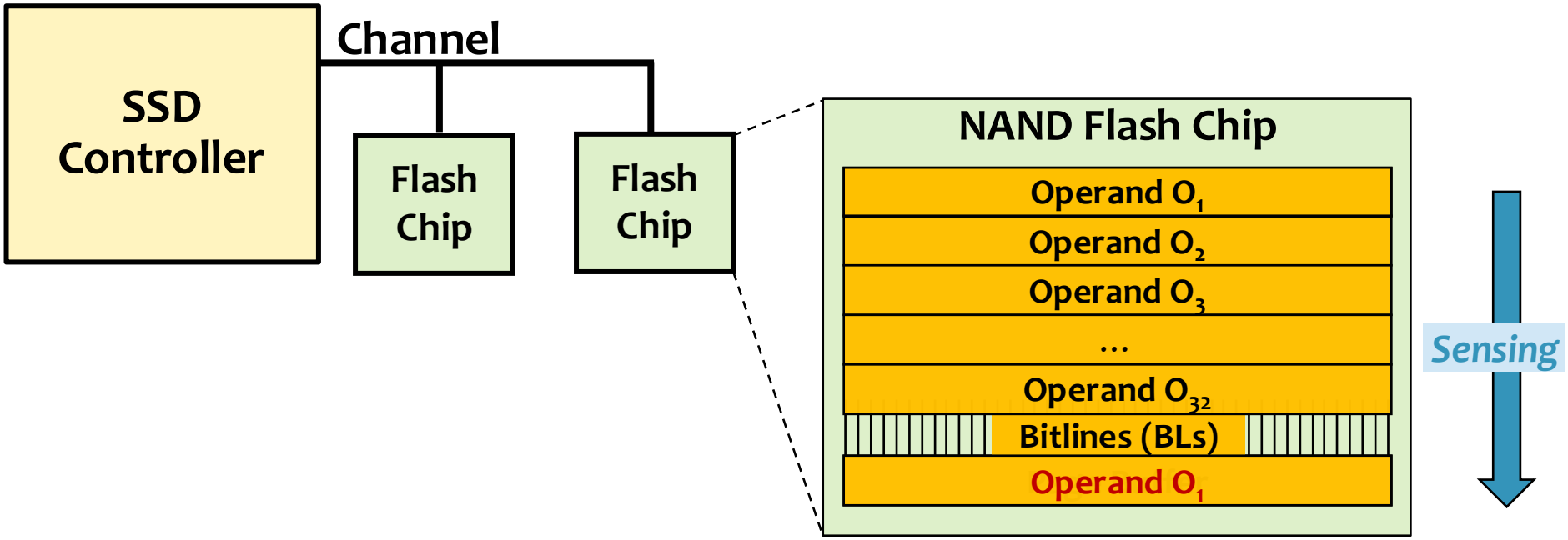
In-Flash Processing (IFP) (1/2)



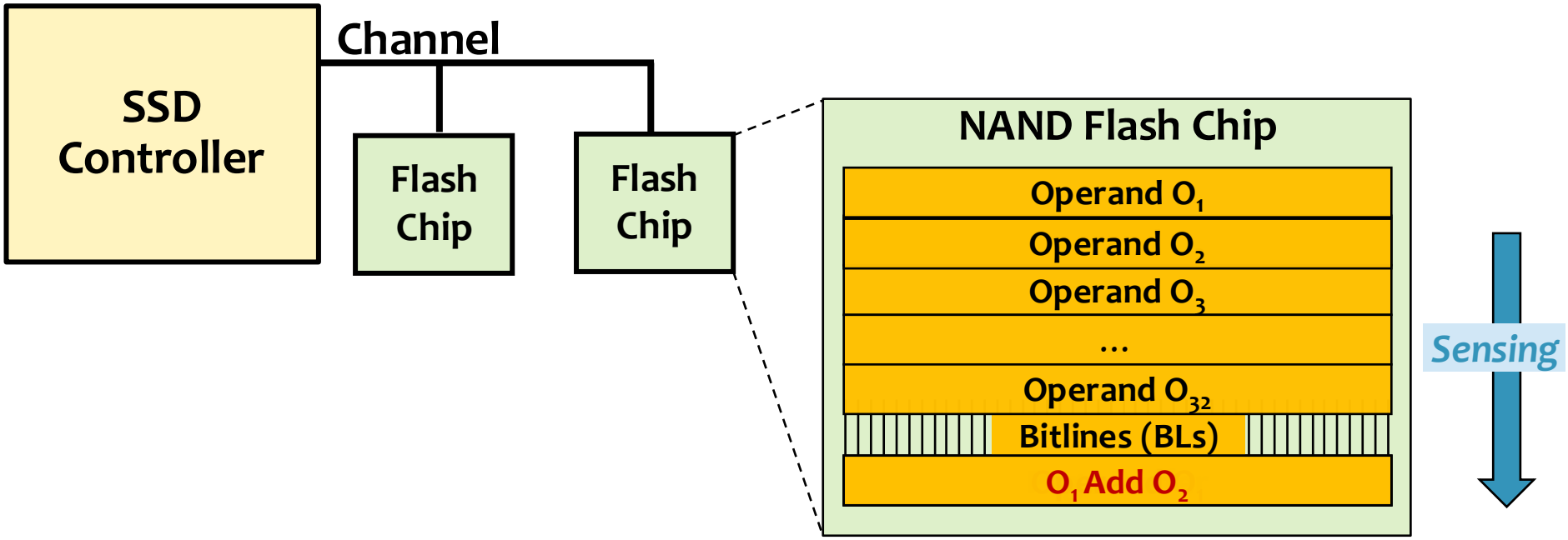
In-Flash Processing (IFP) (1/2)



Several IFP techniques (e.g., Ares-Flash[3]) enable arithmetic operations inside NAND flash chips by controlling the latching circuit of the page buffer

[3] Chen+, "Ares-Flash: Efficient Parallel Integer Arithmetic Operations Using NAND Flash Memory," In MICRO, 2024

In-Flash Processing (IFP) (1/2)



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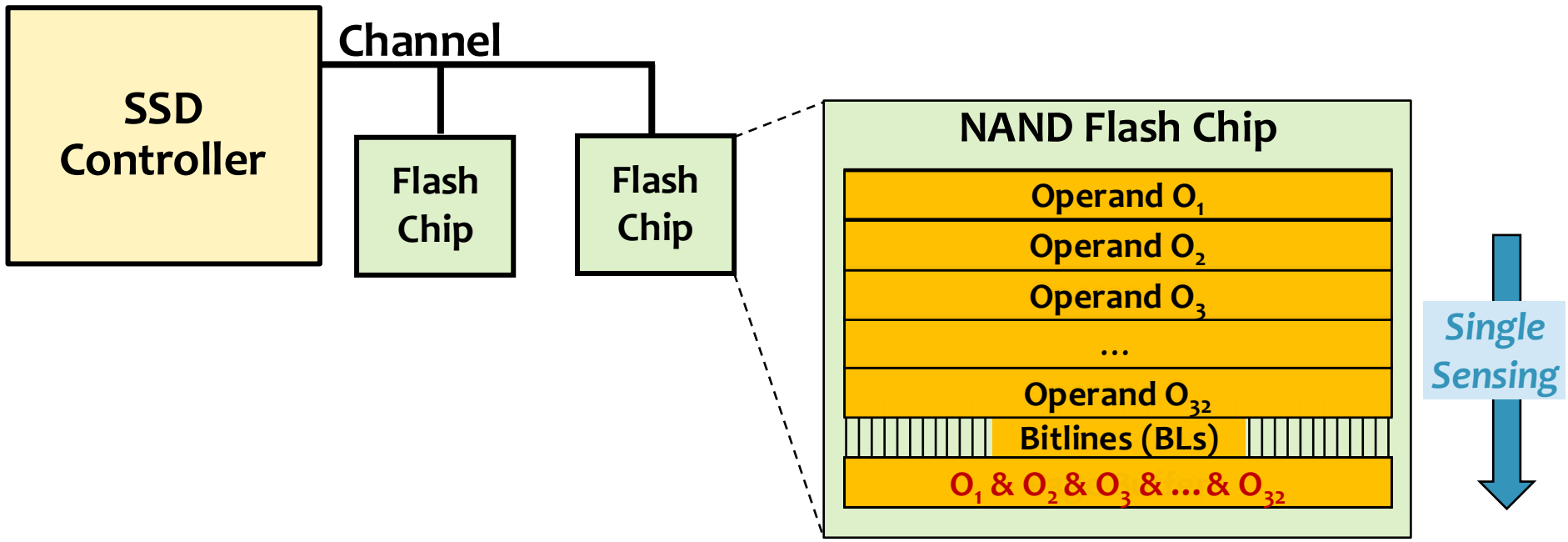
In-Flash Processing (IFP) (2/2)



Flash-Cosmos[4] enables bulk-bitwise operations inside NAND flash chips by simultaneously reading multiple pages within the flash chips

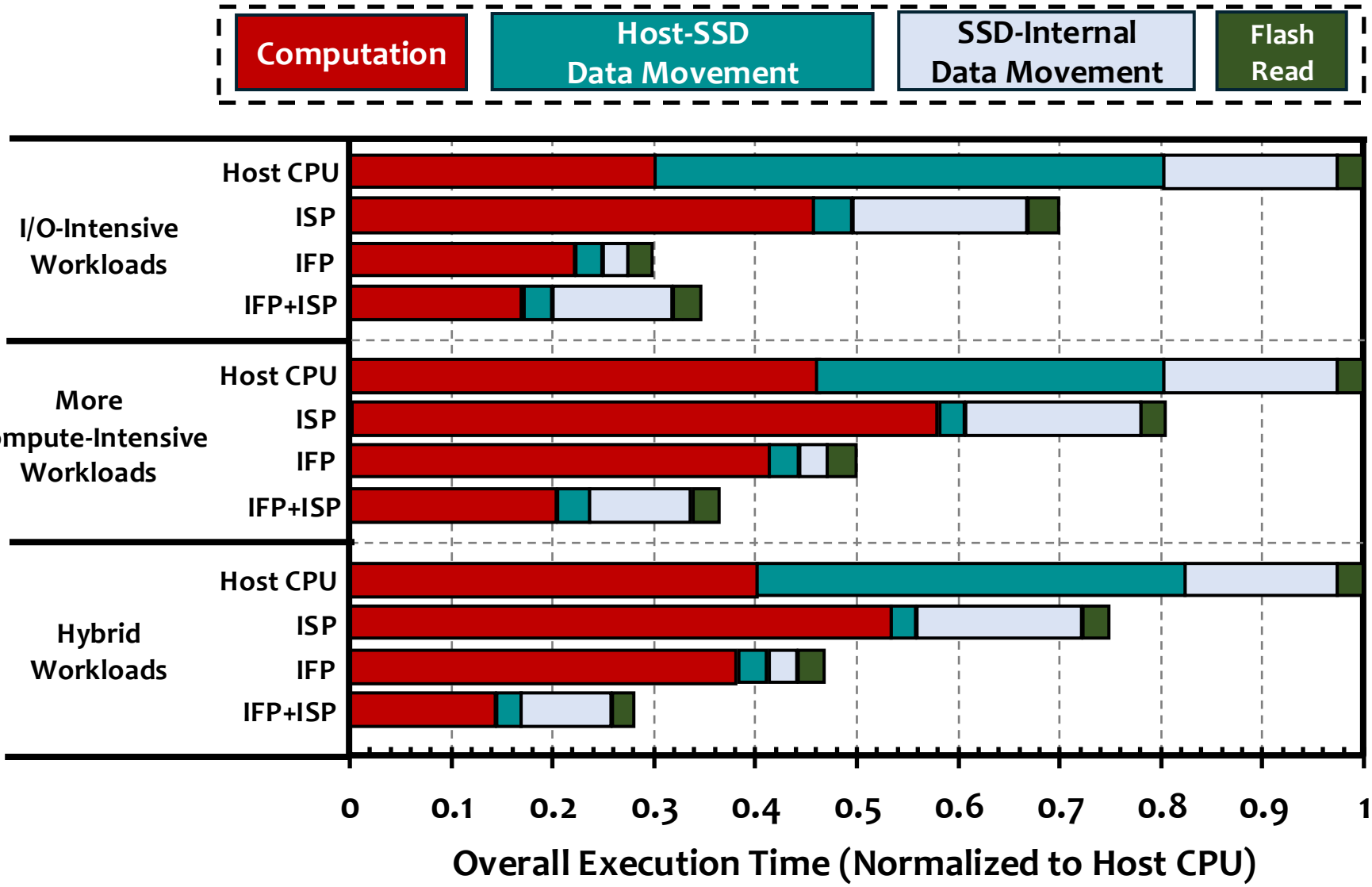
IFP techniques support limited set of operations and are less suitable for complex workloads

[4] Park+, "Flash-Cosmos: In-Flash Bulk Bitwise Operations Using Inherent Computation Capability of NAND Flash Memory." In MICRO, 2022

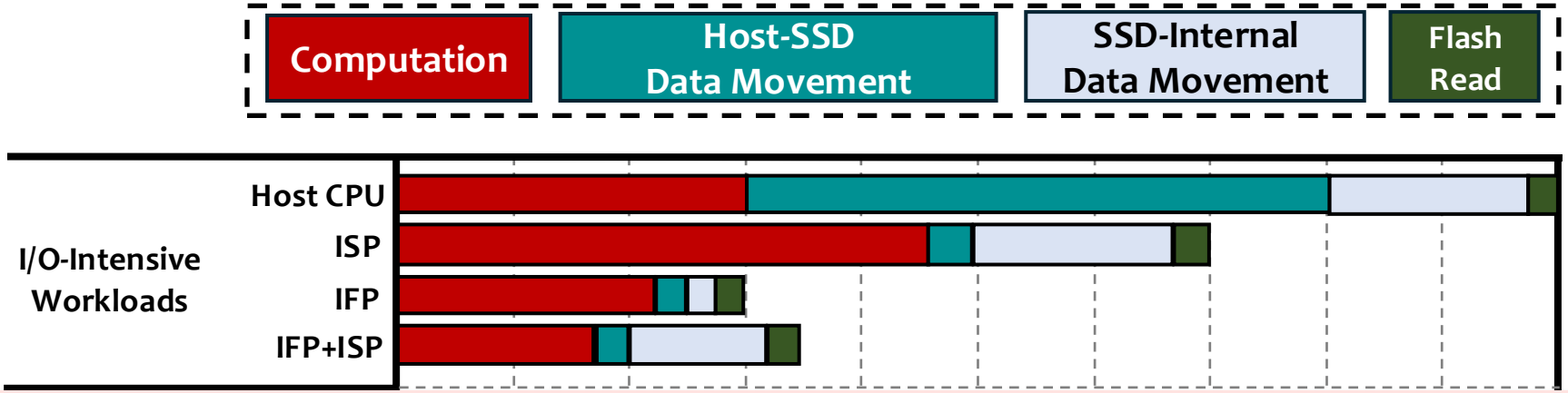
Extended Case Study (1/5)

- We examine the performance bottlenecks when offloading computations to multiple SSD computation resources
- Evaluated Policies
  - Host CPU
  - In-storage processing (ISP)
  - In-flash processing (IFP)
  - Offloading to both ISP and IFP (ISP + IFP)
- Evaluated Workloads
  - I/O-Intensive workloads: Workloads dominated by data transfer (e.g., databases, bitmap indices)
  - More compute-intensive workloads: Workloads dominated by computations (e.g., encryption, matrix multiplication)
  - Hybrid workloads: Workloads that involve both computation and data movement (e.g., aggregation, sorting)

Extended Case Study (2/5)

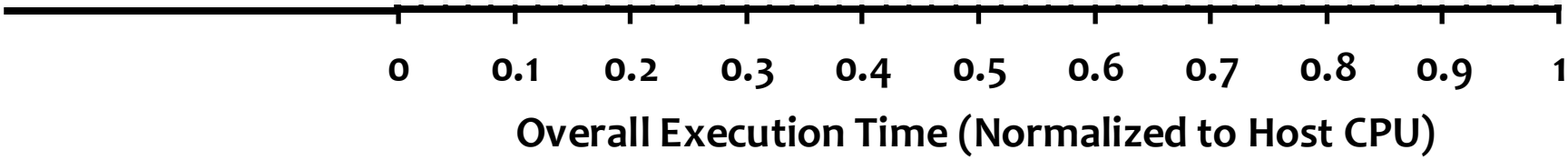


Extended Case Study (3/5)

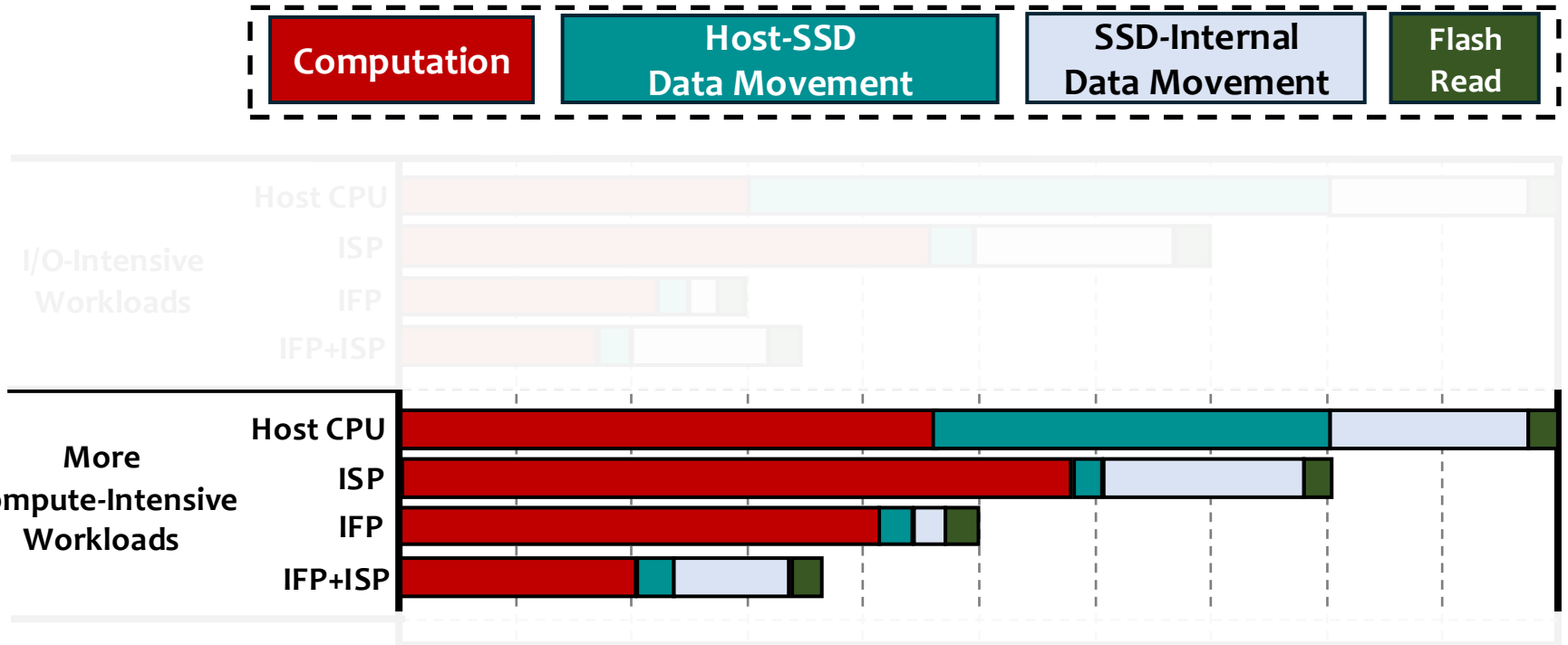


Offloading to ISP and IFP in I/O-intensive workloads reduces performance by 15% compared to IFP

Offloading to multiple computation resources incurs additional data movement between flash chips and SSD controller



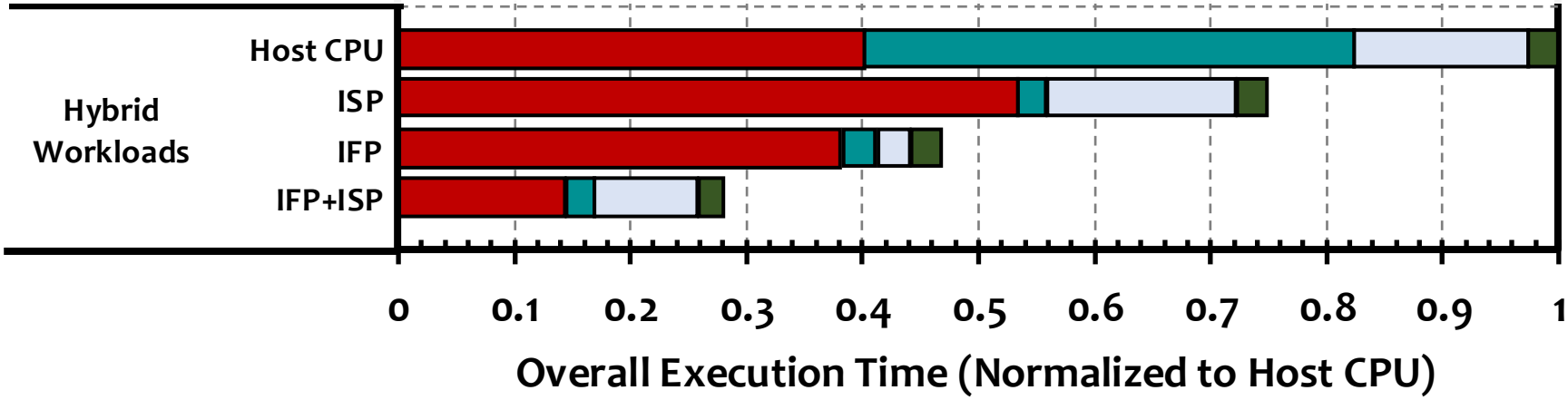
Extended Case Study (4/5)



Offloading to ISP and IFP improves performance by 28% over IFP by mitigating the limitations of individual computation paradigms

Extended Case Study (5/5)

Offloading to ISP and IFP improves performance by 40% over IFP by reducing data movement and mapping computations to the most suitable computation resource



Compile-Time Preprocessing

- **Compiler Flags**
  - -O3 -g –mllvm -force-vector-width=4096 -force-vector-interleave=1 -Rpass-analysis=loop-vectorize -Rpass=loop-vectorize
- -force-vector-width=4096 - Aligns each vector to a typical NAND flash page (vector width of 4096 for 32-bit operands)
  - Simplifies data movement across computation resources
  - Avoids misaligned accesses
  - Matches L2P mapping granularity
- -force-vector-interleave=1 flag controls loop unrolling to enable instruction-granularity offloading

Runtime Offloading: Cost Function

Feature	Description
Operation Type	Type of operation (e.g., bulk-bitwise, arithmetic, predication)
Operand Location	Current location of the operand (flash chips or SSD DRAM)
Data Dependence Delay ($delay_{dd}$)	Delay for operand availability
Resource Queueing Delay ($delay_{queue}$)	Delay caused by pending instructions in a computation resource's execution queue
Data Movement Latency ($latency_{dm}$)	Data movement latencies if the operands are not in the target computation unit
Expected Computation Latency ($latency_{comp}$)	Estimated instruction execution latency on a computation resource

Table 1: Features Used by the Cost Function to Select the SSD Computation Resource.

Runtime Offloading: Why do it in SSD?

- SSD controller has real-time knowledge of all computation resources and shared resource contention
- We explored offloading in the host CPU
- Limitations of offloading in the host CPU
  - No real-time status of the SSD internals in the host
  - Significant traffic if the SSD needs to transfer information to the host frequently
  - We need to think of a better way to do this in the future

System Integration (1/4)

- **Transferring the binary**
  - Repurpose fw-download and fw-commit commands
  - Extend these commands with a new flag to identify Conduit's executable
- **Data mapping and SSD-internal data movement**
  - Conduit adheres to the data layout requirements of prior techniques
- **Host-SSD communication**
  - *Regular I/O mode:* Host I/O and FTL operations are executed
  - *Computation mode:* Host I/O is suspended while performing maintenance tasks such as garbage collection and wear-leveling

System Integration (2/4)

- **Data Coherence between SSD computation resources**
  - Conduit employs a lazy coherence mechanism
  - Conduit synchronizes data across SSD computation resources only when
    - Another computation resource requests data
    - Computation result must be transferred to the host
    - Data must be evicted from SSD DRAM, page buffers or SSD controller registers
    - FTL initiates garbage collection
    - SSD undergoes power cycle

System Integration (3/4)

- **Data Coherence between SSD computation resources**
  - Conduit maintains coherence at page-granularity by extending the logical-to-physical mapping table to store:
    - *Owner* - SSD computation resource that holds the latest version of the page
    - *State* – page's modification status (i.e., clean or dirty)
    - *Version* – 1-byte monotonically-increasing counter to order updates and detect stale copies

System Integration (4/4)

- **Failure Handling**
  - For transient failures (e.g., instruction abort, ECC correction failures), the SSD controller maintains completion and status flags, and replays the instruction in a different computation resource
  - For permanent faults (e.g., bad blocks), Conduit relies on existing FTL recovery mechanisms such as bad-block remapping, garbage collection and wear-leveling

Evaluated Configurations

Host CPU (Real System)	Intel(R) Xeon(R) Gold 5118
	x86-64 [187], 6 cores, out-of-order, 3.2 GHz
	L1 Data + Inst. Private Cache: 32kB, 8-way, 64B line
	L2 Private Cache: 256kB, 4-way, 64B line
	L3 Shared Cache: 8MB, 16-way, 64B line
Host GPU (Real System)	NVIDIA A100 GPU [188], [189], NVIDIA Ampere Architecture [190] (7nm)
	108 Streaming Multiprocessors, 1.4 GHz base clock;
	L2 Cache: 40 MB; Main Memory: 40GB HBM2 [191], [192]
Host Main Memory (Real System)	32GB DDR4-2400, 4 channels, 1 rank, 16 banks;
	Peak throughput: 19.2 GB/s;
	Latency: T_{bbop}: 49 ns; **Energy:** E_{bbop}: 0.864 nJ; where *bbop* is bulk bitwise operation
SSD [87] (Simulated)	48-WL-layer 3D TLC NAND flash-based SSD; 2 TB
	External Bandwidth: PCIe 4.0, 8GB/s
	NAND Config: 8 channels; 8 dies/channel; 2 planes/die; 2,048 blocks/plane; 196 (4×48) WLs/block; 4 KiB/page
	Flash Channel Bandwidth: 1.2 GB/s
	Latency: T_{read} (SLC mode): 22.5 μs [10]; $T_{AND/OR}$: 20 ns [14]; $T_{latchtransfer}$: 20 ns [14]; T_{XOR}: 30 ns [10]; T_{DMA}: 3.3 μs;
	Energy: E_{read} (SLC mode): 20.5μJ/channel [10]; $E_{AND/OR}$: 10nJ/KB [14]; $E_{latchtransfer}$: 10nJ/KB [14]; E_{XOR}: 20nJ/KB [10]; E_{DMA}: 7.656μJ/channel;
SSD DRAM (Simulated)	2GB LPDDR4-1866 DRAM cache; 1 channel, 1 rank, 8 banks
	Latency: T_{bbop}: 49 ns; **Energy:** E_{bbop}: 0.864 nJ; where *bbop* is bulk bitwise operation
SSD Controller Cores	ARM Cortex-R8 series @1.5GHz; 5 Cores [94]

Workloads (1/3)

- **AES**
  - 256-bit encryption and decryption algorithm with high data reuse and low-latency bitwise operations
- **XOR Filter**
  - A probabilistic data structure for fast membership tests
  - Dominated by arithmetic and predication operations
- **heat-3d**
  - 3D stencil computation that repeatedly updates each grid point using its neighbors
  - Moderate-to-high arithmetic intensity with high data reuse
- **jacobi-1d**
  - 1D stencil computation solver that updates each element using values from its immediate neighbors
  - Moderate-to-high latency operations

Workloads (2/3)

- **LlaMA2 Inference**
  - INT8 inference of a 7B-parameter LlaMA2 model
  - High and medium latency operations
- **LLM Training**
  - Bandwidth-intensive INT8 training of 7B-parameter model characterized by frequent weight updates, data movement, and a combination of arithmetic and control-intensive operations
- Quantization enables complete execution of all workloads within the SSD
- Memory footprint of each workload exceeds the SSD capacity by 2x
  - Induces resource contention and data movement scenarios

Workloads (3/3)

Workload	Avg. Reuse	Low Latency Operations	Medium Latency Operations	High Latency Operations	Units
AES	15.2	87%	13%	0%	Single
XOR-Filter	2.0	1%	98%	1%	Single
heat-3D	16	0%	60%	40%	Multiple
jacobi-1D	3	0%	67%	33%	Multiple
LlaMA2 Inference	1.8	0%	53%	47%	Multiple
LLM Training	5.2	0%	88%	12%	Multiple

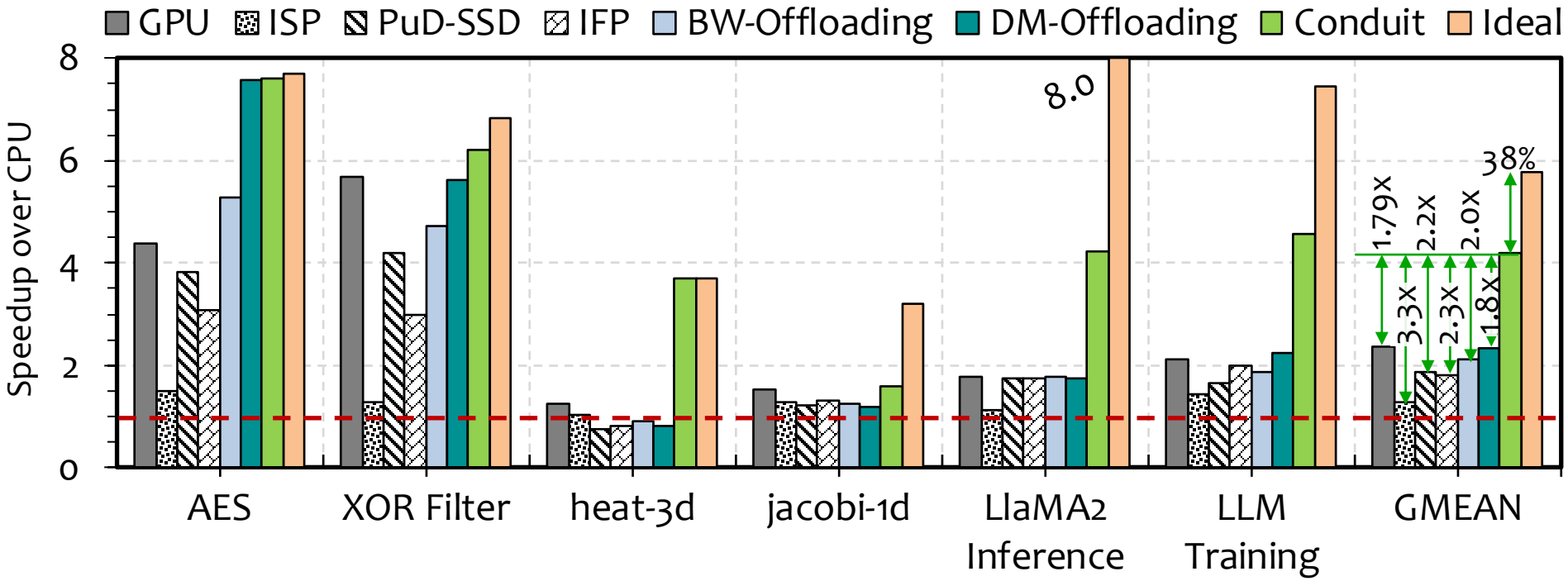
LLM Inference and Training Characteristics

- **LLM Inference**
  - Linear layers (Q, K, V, FFN)
    - Matrix multiplications, high arithmetic intensity
  - Attention-Score computation
    - Dot products
  - Element-wise operations: scaling, residual add…
  - Layer normalization
    - Accumulation, scale and shift
  - Partially vectorizable – reduction, exponentiation, normalization
  - Not vectorizable
    - Token selection and sampling
      - Branch-heavy, data-dependent

LLM Inference and Training Characteristics

- **LLM Training**
  - Forward Pass
    - Vectorizable
  - Backpropagation
    - Gradient computation: Dense matrix computations, high reuse, arithmetic-heavy
  - Weight updates
    - Element-wise arithmetic (SIMD friendly)
  - Optimizer logic
    - Conditionals, state updates and control flow

Extended Results: Performance Analysis



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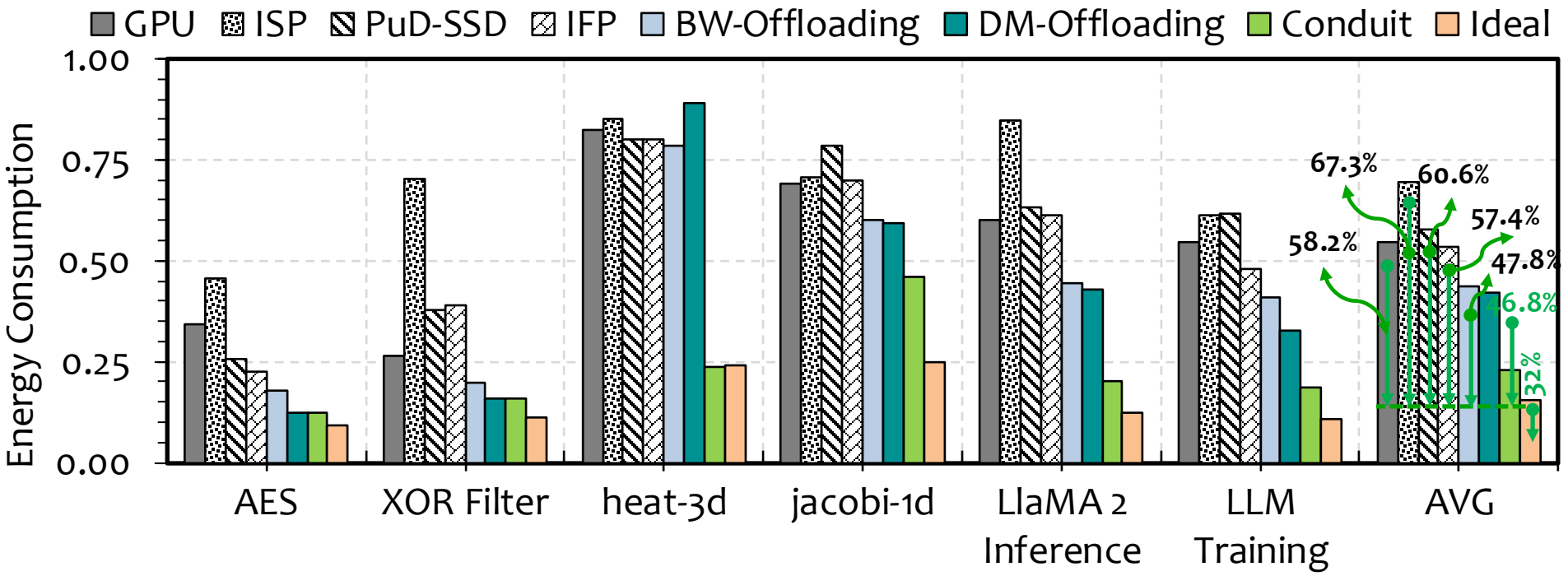
Extended Results: Energy Consumption



Conduit reduces energy consumption by 46.8% on average compared to DM-Offloading by reducing resource queueing delays and exploiting all SSD computation resources

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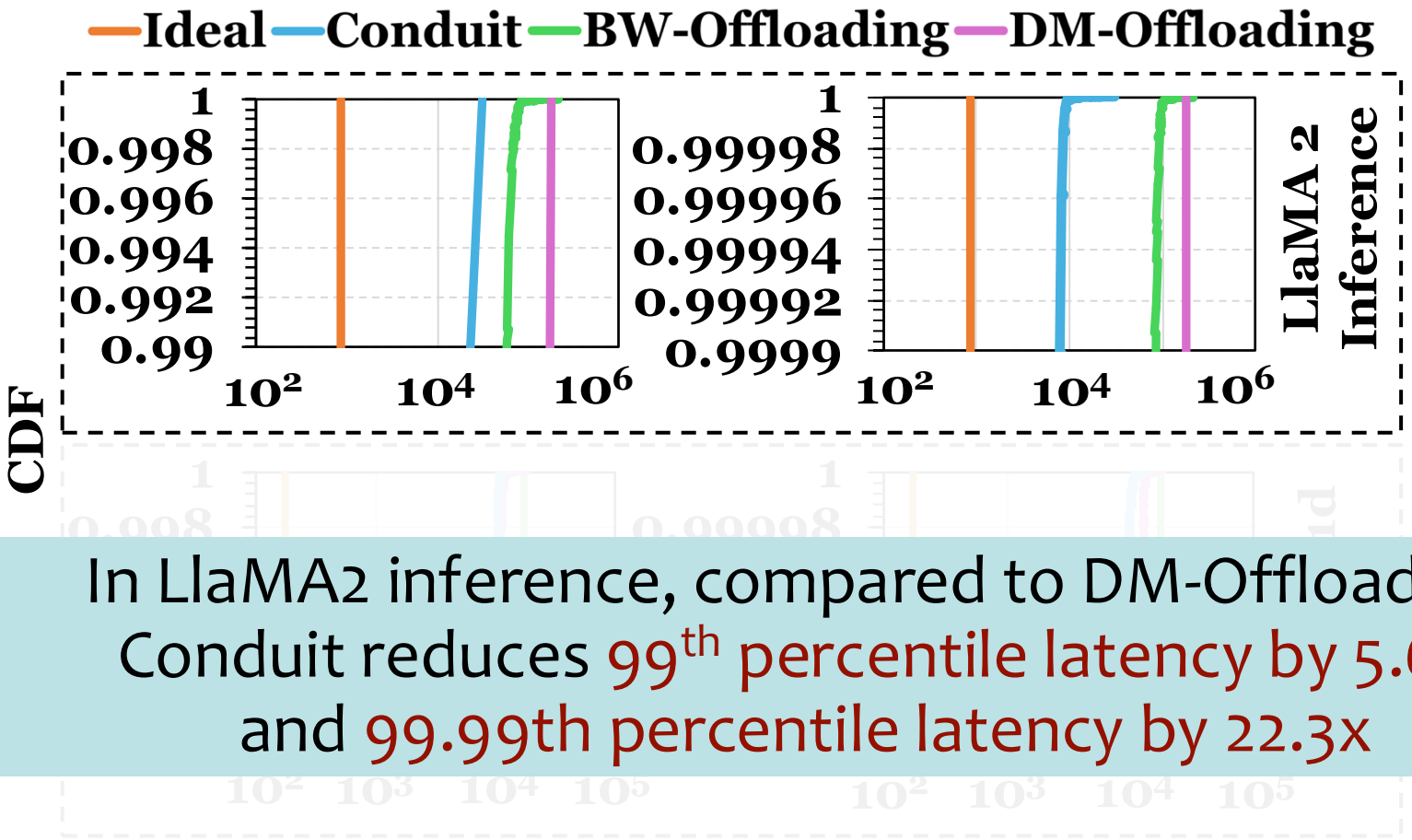
Tail Latency Evaluation (1/2)



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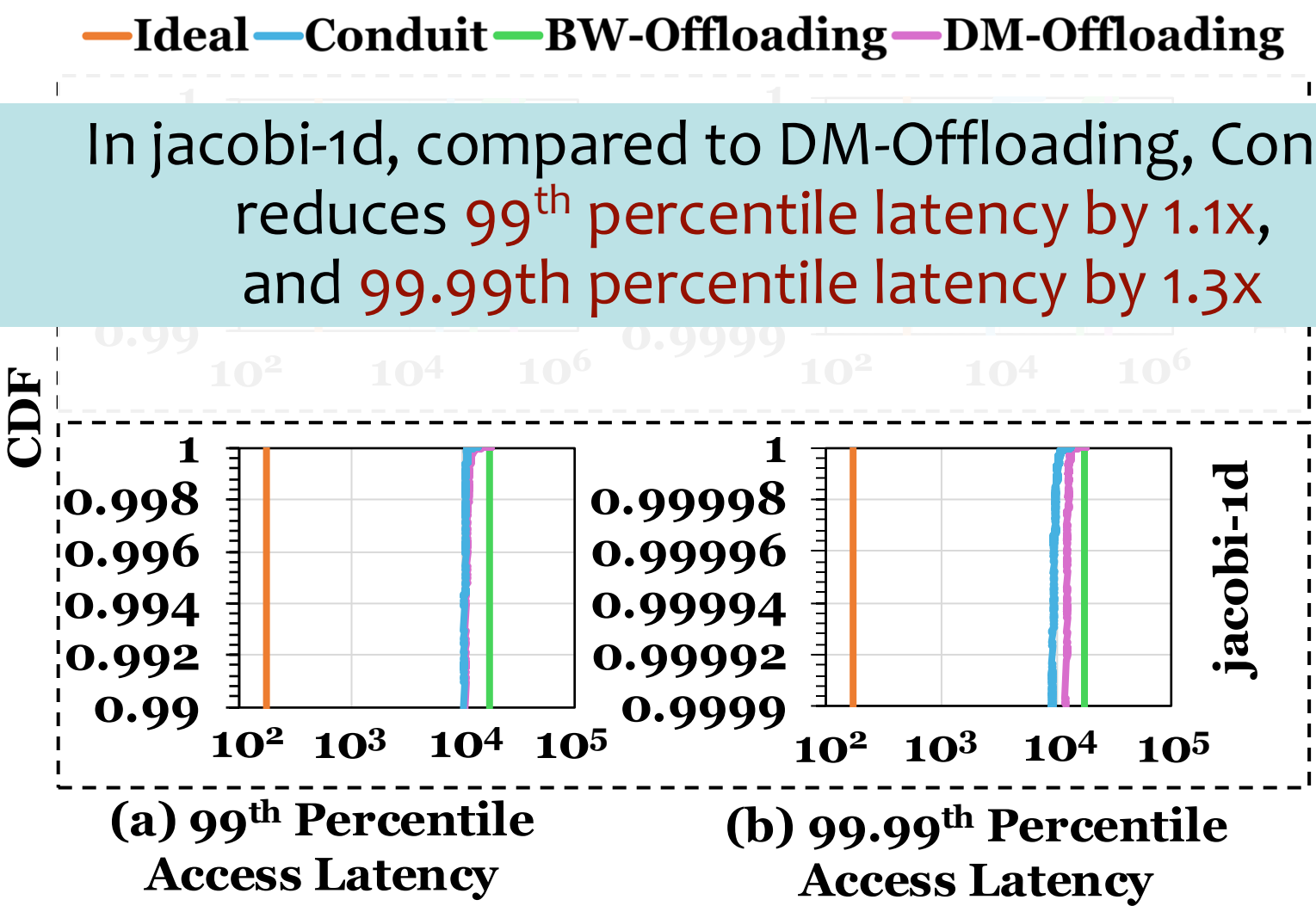
(a) 99th Percentile Access Latency

(b) 99.99th Percentile Access Latency

Tail Latency Evaluation (2/2)



Workload-Computation Resource Interaction